

Trends in biomedical signal feature extraction

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ABSTRACT

Signal analysis involves identifying signal behaviour, extracting linear and non-linear properties, compression or expansion into higher or lower dimensions, and recognizing patterns. Over the last few decades, signal processing has taken notable evolutionary leaps in terms of measurement – from being simple techniques for analysing analog or digital signals in time, frequency or joint time–frequency (TF) domain, to being complex techniques for analysis and interpretation in a higher dimensional domain. The intention behind this is simple – robust and efficient feature extraction; i.e. to identify specific signal markers or properties exhibited in one event, and use them to distinguish from characteristics exhibited in another event. The objective of our study is to give the reader a bird's eye view of the biomedical signal processing world with a zoomed-in perspective of feature extraction methodologies which form the basis of machine learning and hence, artificial intelligence. We delve into the vast world of feature extraction going across the evolutionary chain starting with basic A-to-D conversion, to domain transformations, to sparse signal representations and compressive sensing. It should be noted that in this manuscript we have attempted to explain key biomedical signal feature extraction methods in simpler fashion without detailing over mathematical representations. Additionally we have briefly touched upon the aspects of curse and blessings of signal dimensionality which would finally help us in determining the best combination of signal processing methods which could yield an efficient feature extractor. In other words, similar to how the laws of science behind some common engineering techniques are explained, in this review study we have attempted to postulate an approach towards a meaningful explanation behind those methods in developing a convincing and explainable reason as to which feature extraction method is suitable for a given biomedical signal.

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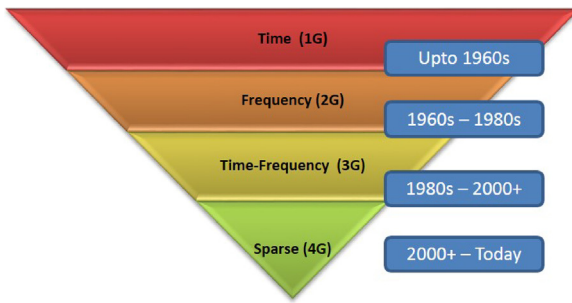


Fig. 1. Evolution of biomedical signal feature extraction.

1. Introduction

Signals are omnipresent. This statement certainly holds true when we are able to represent most stationary and non-stationary phenomenon in mathematical expressions. These representations are able to give us keen insights into those phenomena, and help us in identifying characteristic patterns of interest. Signal processing involves analysing analog/digital signals with the intention of measurement, reconstruction, quality improvement, compression, feature extraction and pattern recognition. Advancements in sensor technologies have come a long way by making signal data acquisition, storage and analysis easier, as well as opening doors for further improvisation considering unstructured big data (Fig. 1).

Someone might ask “If signal analysis should be easier to achieve with developments in signal processing algorithms, then how come we have to deal with increasingly complex mathematical representations and optimization problems?” The only suitable answer to this would be that – modern day information theory relies extensively on big data signals being churned out by sensors from our natural and digital environments, and treating these signals requires algorithms that are highly efficient in storage and computation. Although the underlying algorithms constitute complex mathematical operations, the flow of code is designed with the intention of processing maximum amount of signal data and discovering characteristic patterns in shortest time possible. This would be conducive only if we are able to seamlessly stream and process data. This in turn motivates us to design better tools for capturing useful information from signals at the source, and discarding unwanted signal artefacts, which would also lead to hardware optimization. As a quick remark, we would like to highlight that this concept has been successfully implemented in state-of-the-art compressive sensing techniques for signal acquisition, analysis and reconstruction.

1.1. Evolution of feature extraction methods

In simple terms, feature extraction is the process of unveiling hidden characteristic information about the input signal and its behavior of its sources. That is, we are able to represent a given input signal by a set of features which represent a specific behavior or pattern depicted by the signal; or a compact or useful representation of the signal [1–5]. Feature extraction is usually a dimensionality reduction or data compression/reduction process and helps in reducing the number of resources required to analyze an input signal. In other words, given a large input signal with multiple redundant components, performing feature extraction on it would yield a smaller set of representative data which could describe the original signal with sufficient accuracy and also help in building an efficient and robust pattern classifier system [1,6–10].

We suggest that the user derives application-dependent features rather than generic features, as they would better suit and depict signal behavior and underlying patterns. For example, when

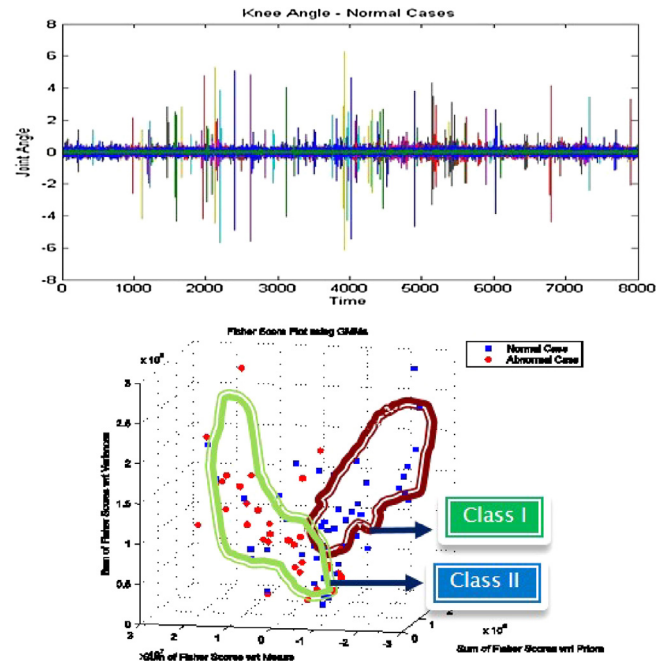


Fig. 2. Example of pattern classification.

we are attempting to analyze and classify music signals, we do not need mean or variance of the signals or even its root mean squares (RMS), since the room or environment settings will be tuned accordingly, but we might need to use them when audio signals are taken from non-modifiable sources. Before we proceed with extracting information from signals, we usually discretize the continuous analog signals into discrete digital signals using an A-to-D converter. This helps in identifying characteristic patterns over discrete time-intervals which otherwise cannot be observed if the signal is processed in analog form (Fig. 2).

At grass roots level, the easiest way to analyze time-domain signals is by filtering them, which helps in removing unwanted artefacts from the signals such as overlapping noise, third party/source components or values, and unwanted signal patterns. The most appropriate method for signal pre-processing method will be the one that can produce an output most suited to feature extraction. This method can be devised through two possible approaches: (a) if the artifact characteristics (such as noise patterns) are known, we can design appropriate signal filters, or (b) if the artifact properties are unknown, we need to pre-process the signal using trial and error approaches. Let us review some feature extraction methodologies applied to real-world biomedical signals in the past few decades. To make it simpler for the reader, let us group all the available signal processing and feature extraction techniques into the following four generations:

- (1) Time domain
- (2) Frequency domain
- (3) Joint time–frequency domain
- (4) Signal decomposition and sparse domains

The reader may note that the list of methods included in our review is by no means exhaustive, and that we have studied some key feature extraction methods in biomedical signal processing, and have attempted to find out the most efficient method from each generation. This study will further define the criteria to design an intelligent feature extractor specific for biomedical signals. In order to better explain and demonstrate our views on various feature extraction techniques, we have running examples

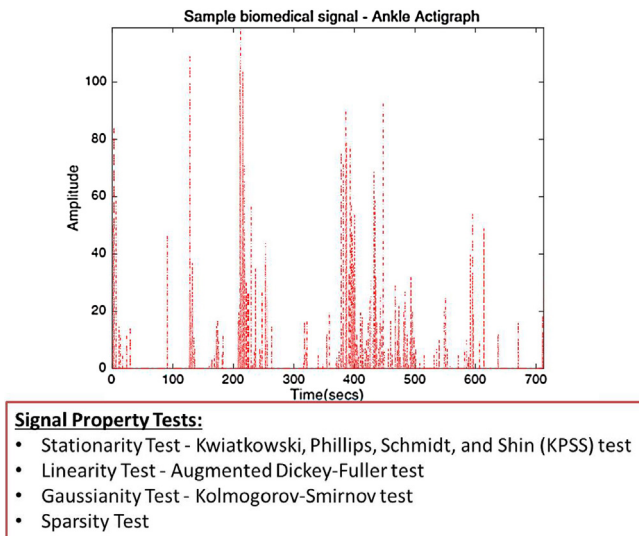


Fig. 3. Signal property tests.

in Sections 2–5. Also, it may be noted that before we apply feature extraction techniques, we must also perform the following set of signal properties' tests, which would lend itself in applying the appropriate mathematical and statistical methods: (i) Non-stationarity, (ii) Non-Linearity, (iii) Normality, (iv) Sparsity, and (v) Multi-component aspects (Fig. 3).

This review work has been organised as follows: Sections 2–5 will briefly describe key feature extraction methods and their efficiencies in characterizing signals. Following this, we will discuss the applications and benefits of some machine learning tools in conjunction with feature extraction methods in Section 6. Further to this, in Section 7 we will highlight some areas concerning the curse and blessing of dimensionality, and how it affects feature extraction and machine learning. Finally, we will conclude our review with some critical discussions, observations and future works in Section 8.

2. Time domain feature extraction

Starting with the basics, we explore simple time-domain signal processing techniques, wherein analog/digital signals are analyzed over time. The visualization would tell us how the signal values change over time, and how to use those values for predictions and regression analysis. Signal processing in time-domain usually involves extracting characteristic properties or features from a specific time window containing say N discrete-time samples [11–13]. The time window can be randomly selected, considering that most biomedical signals we encounter are non-linear and non-stationary, but the underlying patterns and properties could remain the same for a specific phenomenon exhibited by the signal's source. At the ground level, the most basic features which could be extracted from the signal would be its **statistical properties** [13,12,14] such as mean and standard deviation (variance). These generic features are generally applicable when we are trying to classify or recognize commonly occurring patterns in signals. This being said, there are various application-specific feature extraction techniques in time domain such as Autoregressive (AR) Modelling/Linear Predictive Coding (LPC), Cepstrum Analysis and Kernel-based modelling.

Of these, AR modeling and LPC revolve around the same idea: “the future values of a discretized signal is calculated as a function of current and previous values”; while Cepstrum analysis propagates the idea of rate of change of different frequency spectrum

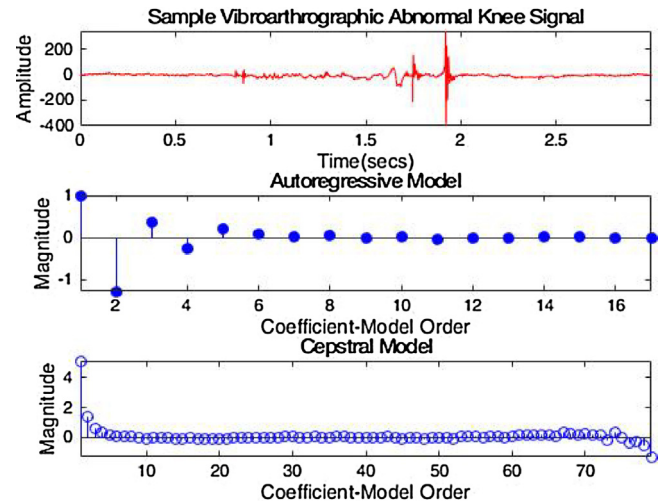


Fig. 4. (A) Sample knee signal, (B) Autoregressive model, (C) Cepstral model.

bands of the signal. It could also be interpreted as homomorphic filtering, wherein the signals have been transformed by joint addition and multiplication operations [15,16]. AR modelling helps in enhancing data compaction, signal resolution, enhancing spectral peaks, and also reduces signal noise [17–21], as shown in Fig. 4(B). But this method has a downside as well – the model order cannot be determined a priori and needs to be optimized within the analysis time window [17–21]. Sometimes if the model order is too small, the main statistical properties of the original signal might get ignored, and if the order is too big, it might result in including additional noise as a result of over-fitting. Also one must note that AR modeling is applicable only to a stationary window selected from a non-stationary signal – which means that continuous feature extraction is possible only if we adaptively segment the non-stationary signal into specific sized windows. AR modeling has been extensively applied in biomedical signal analysis including – cell and tissue characterization, biometric modeling, EOG analysis, EMG signal analysis, characterizing knee-joint signals or gait analysis, bioacoustic signals [15–21].

Cepstrum analysis was initially applied to analysing seismic echoes from earthquakes and other geo-physical signals [22]. The technique was then extended to processing radar signals and human speech analysis [23,24,16,11], and has proven to be effective in discriminating between human sounds. It may be noted here that the Cepstrum coefficients calculation does not require computing a Fourier transform, and hence it could be considered as time-domain feature. Also it must be noted that Cepstrum modeling requires that the signal to be analysed must be stationary over a given time interval. But as we know that real-world signals cannot be stationary, a workaround to use these time-domain techniques is to perform adaptive segmentation on the signal before feature extraction. For representing human speech signals we mainly use the power Cepstrum as a feature vector which leads us to an improvised set of features known as MFCCs or Mel Frequency Cepstral Coefficients [25]. These features are calculated by transforming the spectrum into Mel scale, thus creating Mel-frequency Cepstrum. These MFCCs efficiently capture spectral energy measurements over short time windows, but tend to lose non-stationary spectral values such as time-varying or transient patterns in the signal [26]. These features are highly useful in current state-of-the-art applications in voice recognition, pitch detection, speaker recognition, speech-based emotion recognition, and pathological voice analysis [27–29].

In time-domain signal processing, LPC has given good results in audio and speech signal compression [12,14]. In terms of feature

Table 1
Summary of time-domain feature extraction methods.

Method	Advantages	Disadvantages	Sample applications
AR Modelling	Signal compression Improve resolution Model spectral peaks Reduce noise	Model order cannot be found a priori Lower order ignores statistical properties Higher order adds noise Only applicable to stationary windows	Cell and tissue characterization Biometric modelling EOG, EMG analysis Gait analysis Audio signal analysis
Cepstrum Analysis	No need to compute Fourier transform Features capture spectral energy	Cannot capture transient patterns in the signal Only applicable to stationary windows	Pathological voice analysis Emotion recognition Speaker recognition
Linear Predictive Coding (LPC)	Features validate signal value prediction Robust to noise and signal quantization Good for signal encoding	Sensitive to additive noise and error Cannot separate overlapping peaks	Cancer Cell analysis Speech encoding
Morphology feature extraction	Accounts for physiological properties Easy to compute features Account for signal structure and hidden information	Cannot capture transient features Only applicable to small signal window Cannot address signal noise and artefacts	Digital histopathology Gait analysis Detect periodic limb movements
Kernel-based Modelling	Reduce signal complexity and dimensions Enable feature visualization Usually linear separation of classes	Might incur complex computations Non-linear transforms may be complex Signals need to be free from artefacts	Gait analysis Speech recognition Financial time-series analysis

extraction, the features extracted from LPC process are the linear prediction coefficients which help in validating if the signal values have been correctly predicted. Depending on a specific signal processing application, a linear prediction coder/filter can be customized to generate suitable coefficients or parameters which can be characterized as features for a particular signal. For example, when we investigate into analyzing structural changes in cancer cells [19], we apply a Lattice prediction filter based on LPC theory to generate reflection coefficients as our features. Linear prediction filters have proven to be good signal processing tools as they are relatively robust to noise and signal quantization effects [19,30].

When analyzing trends and patterns in biomedical signals, we must always consider two important aspects: (a) structure and dynamics of the physiological system emanating the signal, and (b) signal properties reflecting the morphology of physiological system. These aspects help us understand some of the key characteristics of physiological or biomedical signals, which could be further extracted as morphology features for pattern classification, disease identification and severity categorization. Our literature survey indicates that rather than specific methods, morphological feature extraction is simply an offshoot of time domain signal analysis techniques, built using a combination of basic mathematical functions and algorithms. For example, morphological feature extraction has been applied in classification of diseases from images captured using digital histopathology [31–34] using features based on shape and sizes of cells. Similarly, periodicity index of limb movements in sleep is computed as a movement-based morphological feature from accelerometry signals for identifying the severity of Periodic Limb Movements (PLM) [35].

Deriving linear relationships within signal components is a desirable approach for developing machine learning tools, so that we can extract suitable features from the input data for the learning algorithm. However, when the input signal exhibits properties of non-linearity and non-stationarity, simple signal processing and machine learning tools tend to produce non-linear separation of classes or patterns exhibited by a group of similar signals emanated from similar sources. For example, neuromuscular activities detected using accelerometry techniques are highly likely to exhibit random variations and differences when captured from different test subjects. This nullifies the possibility of applying simple signal processing tools for pattern classification. One solution to address this problem is the application of kernel methods to the

input signal, which would transform it into a higher dimensional space, with a compact representation, and where it would be easier to distinguish between signal classes.

In kernel-based time series modelling methodologies, complex real-world signals undergo dimension transformation, usually from a single-dimension time series signal to a collection of multi-dimension mapped points in an n – dimensional space [1,36,37,20]. The aim of a kernel function is to reduce the complex signal to simple set of features or representative values in (usually) higher dimensional space, so that identifying patterns (mostly through visualization) becomes easier. Although the kernel mapping might be non-linear in nature, the inter-class separation itself would be linear. Kernel based signal modelling methods have been extensively used in analyzing financial time series [1,36], biomedical time series such as gait signals [36], speech and speaker recognition [38–41].

In order to better explain time domain signal analysis, we have taken an example of a vibroarthrographic abnormal knee signal acquired from a test subject [17], and extracted a small set of features or coefficients using Autoregressive modelling and Cepstral analysis, as illustrated in Fig. 4.

Although time-domain analysis of physiological signals may yield us significant statistical and morphological features, it sometimes would not be able to represent a signal accurately due to presence of certain artifacts or complex behaviour, which could be addressed only through the application of complex methodologies such as mathematical transformations to another domain. Nevertheless, we still need to understand the evolutionary chain and hence the root methods of feature extraction before we delve into complex methodologies. It may be noted here that time-domain techniques will always reveal core information about signal properties which would always prove helpful in developing or deriving higher dimensional features for pattern analysis. Of the various methods discussed in this section, AR models, Cepstral analysis and kernel methods have proved to be widely applied time-domain based feature extraction methods (see Table 1).

3. Frequency domain feature extraction

In most modern day signal processing methods, frequency domain transformation and feature extraction has become a quintessential aspect of signal processing. In simple words, a time-

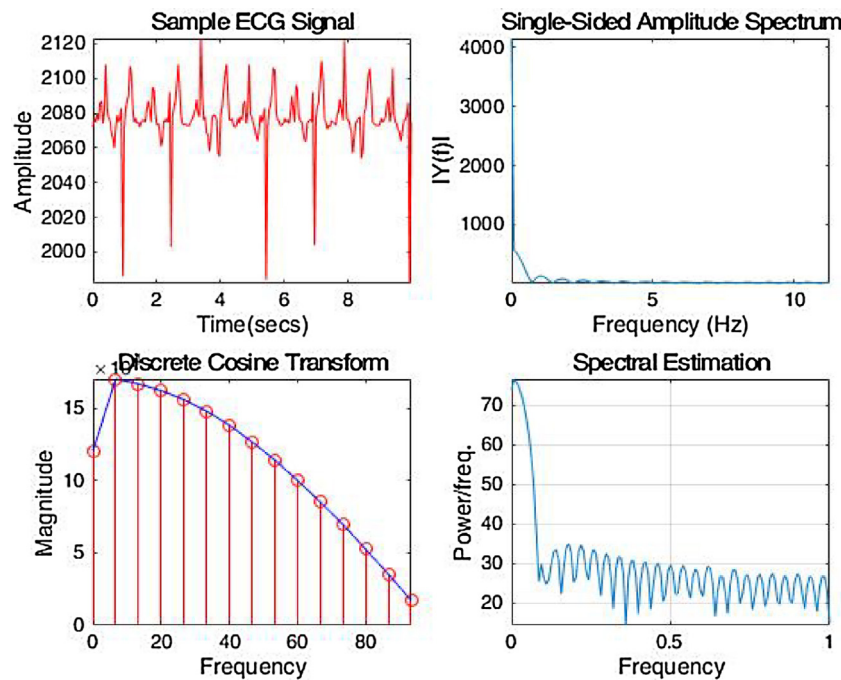


Fig. 5. (A) Sample ECG signal, (B) Magnitude Spectrum, (C) DCT, (D) spectral estimate – Welch method.

domain signal tells us how the real-world signal varies with time, whereas a frequency domain signal indicates the rate of change in signal values and its spectral composition. The most common transformation used in biomedical signal analysis [12,11,42,43] is the Fourier transform – which converts any given practical signal (time limited) into a sum of infinite number of sinusoidal waves. Most notable contribution of using frequency transforms is aid they provide in identifying the artifact frequencies masking vital signal information, which could then be used to develop suitable filters for noise removal.

Most of the signal processing applications require only the frequency information for feature extraction and pattern classification; although some do use the phase information [42,43]. Most importantly, a Fourier transform gives us information about what frequencies are present in our signal and in what ratios. Transformation from time domain to frequency domain helps us understand that convolution in time domain is equivalent of multiplication in frequency domain, and vice versa. Fourier transforms can help the user break a real-world signal into meaningful components for pattern recognition and classification [11]. Fourier transforms are applicable to both discrete and continuous signals, as long as they are integrable – which might also lead the reader to explore Laplace Transforms. The Fourier analysis domain is a vertical comprising of numerous methods which are derived from the basic Fourier Transform.

One of the earliest variations is the Discrete Fourier Transform (DFT) which converts a finitely sampled signal from time domain to frequency domain. DFT differs from DTFT (Discrete-time Fourier Transform) in the sense that both input and output signals are finite in DFT. Since DFT deals with finitely sized data, it can easily be implemented on signal processing hardware. These systems usually implement FFT (Fast Fourier Transform) which is a faster implementation of DFT [14,44]. DFT is computed by applying the FFT (Fast Fourier Transform) algorithm, which rapidly computes Fourier transforms by factorizing the Fourier transform matrix into a product of sparse or small number of significant factors – which could be considered as features for machine learning and subsequent pattern classification (other features include Eigen values

or Eigen vectors). This leads to applying DFT/FFT into numerous applications in biomedical signal processing such as ECG analysis [45], audio spectral analysis [46,47], data compression and multi-channel EEG filtering [48], and also in wireless networks for signal separation [14].

In case of multi-channel signals, retaining original signal is often recommended so that reverting to time-domain for changing areas/samples of interest could be done conveniently. This implies that DFT/FFT tends to discard the non-selected time-domain samples during transformation, thus leading to the risk of loss of signal values/information. A major disadvantage of DFT is that it does not work well with non-stationary signals and fails to capture spectral variations [44]. Thus DFT also fails in capturing instantaneous frequency information in non-stationary signals – which can be very well be captured by joint time–frequency methods. Postulation and application of DFT has also led to developing other novel variants of Fourier transform such as STFT (Short-time Fourier Transform) [49] which have also proved to be effective in solving signal analysis problems [25].

Similar to DFT, the Discrete Cosine Transform (DCT) is another widely used frequency domain feature extraction technique, which converts a discrete time domain signal into a sum of cosine functions (whose coefficients are our features) which oscillate at different frequencies in the domain [42,14]. DCT is very similar to DFT, except that the cosine transform is applied only on real numbers and DCT uses only cosine functions for transformations. DCT is widely used for signal compression applications (audio and images) because of its energy compaction property. This emanates from its fundamental property – converting data points into a sum of cosine functions at different oscillations, thus making the transformation orthogonal in nature [14]. A major disadvantage of DCT is that although the input values could be integers, the output will always be real-valued. In order to maintain integrity, we need some quantization step to make the output integer-valued [50].

A good way to assess the frequency content of a signal is to monitor its spectral density at different points of interest. This also helps us in finding periodicities from a short window in an otherwise complicated signal. Spectral estimation techniques could

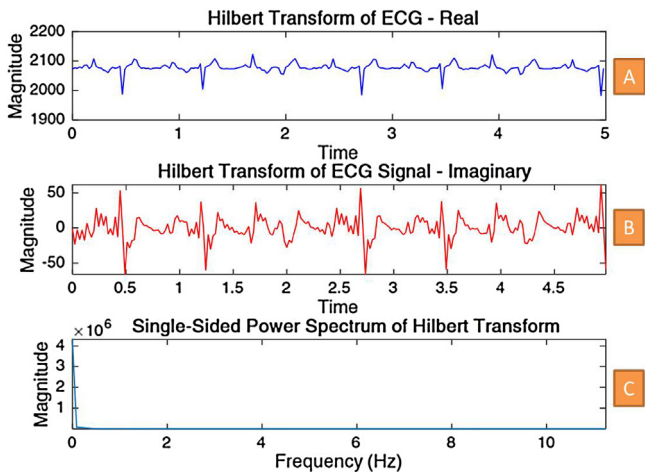


Fig. 6. (A) Real part of Hilbert transform, (B) imaginary part of Hilbert transform, (C) power spectrum.

be categorized into parametric and non-parametric [19,21,17]. The parametric approach works under the assumption that the input single exhibits a specific pattern described by its statistical features derived using methods such as AR-modelling. On the other hand the non-parametric approach explicitly calculates the spectrum of the signal without considering its structure or properties. Based on the spectral density estimation various methods such as Bartlett's method, Welch's method [44] have been developed which employ variations of the conventional Periodogram method for estimating spectral density parameters of a frequency-domain signal. Despite their robustness to noise and quantization effect, spectral estimation techniques fall short when it comes to estimating densities of instantaneous frequency components as they are applied directly to a finite signal window with an averaging effect. Nevertheless these techniques have been applied in audio signal processing for music classification, speaker recognition [46], in biomedical signal processing such as ECG analysis [26,51,52], and in image fingerprinting and watermarking applications [53–55].

In order to better highlight key frequency domain methods, we have used a sample ECG signal [56] as illustrated in Fig. 5. The reason behind using ECG as an example is that its periodic nature allows us to identify significant frequency features easily.

When it comes to analysing phase information and extracting features in frequency domain, Hilbert transform is widely applied as it not only gives us instantaneous components but also preserves signal energy. Hilbert transforms define relationships between real and imaginary parts of complex signals [24,57]. Unlike spectral estimation techniques, Hilbert transform performs really well when extracting instantaneous time and frequency domain features such as amplitude, phase and dominant frequency. Since Hilbert transform is not a bounded operation, it may need certain additional operations before implementing it on discrete signals. Despite this, Hilbert transform has been extensively applied in calculating instantaneous features of ECG signals during ventricular fibrillation, EEG signals [24], filter designing, sampling bandpass signals for communication and AM-FM decomposition in auditory prostheses [57]. Fig. 6 illustrates the application of the Hilbert transform on a 10-s ECG signal [56].

Transforming to the frequency domain not only helps in better signal visualization and interpretation, but also gives a local framework for low-level feature extraction and signal classification. These methods are the building blocks for developing methods which transform a modern non-stationary complex signal into higher dimensions (such as joint time–frequency plane) for extracting characteristic and hidden information. For example, the Fourier

transform is a quintessential method when transforming a signal to a spectral domain; the DCT is used for building dictionary of matching functions, which further help in signal size reduction through sparse representation; techniques such as DFT, and spectral estimation provide us a good basis for biomedical image analysis and 2D signal analysis, and works best with windowed signals.

We know that Fourier transform is widely used in most signal processing applications, but its cosine counterpart – the DCT enhances signal energy conservation and feature retention. The DCT feature extraction leads to exploiting the property of data compression, which in turn helps in hardware optimization of transmitter-receiver sensors, for improved signal analysis of real-world signals. A key highlight of any frequency domain technique transformation is the resolution change it imposes on the signal, which brings out the hidden information to extractable level, as well as enhances signal visualization, especially when analyzing sharp discontinuities in signal behavior. Table 2 summarizes key aspects and applications of various frequency-domain feature extraction methods.

4. Joint time–frequency domain feature extraction

Through time-domain and frequency-domain techniques we can usually extract low-level features from a windowed signal. But these features do not represent the true non-stationarity of real world signals and capture only the global information which roughly classifies the signals. Real world signals are non-linear and non-stationary, and in most cases their characteristic information lies in transient and localized components which can be analyzed only by transforming them to suitable dimensions. One way to do this is to apply time–frequency transformation to a non-stationary signal [4,58,28,59]. By applying suitable processing on the TF decomposition parameters, even subtle signal characteristics can be revealed. In many real world applications, identification of these subtle differences makes a significant impact in signal analysis. Particularly in classification applications using TF approaches, there may be situations where a localized highly discriminative signal structure is diluted due to the presence of other overlapping signal structures. When we apply TF representations to signal classification, we observe that there are small regions where multiple signal components overlap with varying discriminative characteristics. The power of any overlapping area is usually determined by the high energy signal components which mask the discriminative characteristics of the less energy components [60–63].

Majority of non-stationary biomedical signals are composed of a mixture of coherent and non-coherent signal structures with varying localized overlapping regions. These overlapping regions can be separated in order for understanding of the signal for extracting high discriminative features. Coherent signal components have definite TF localization and hence modeling and correlating them with dictionary elements is easier through greedy search methods such as matching pursuit algorithms. On the other hand, the non-coherent components need to be broken into smaller structures till their information is diluted across the whole dictionary. Composition of coherent and non-coherent structures in a signal decides the energy distribution pattern in the signal and hence the decomposition algorithms [60–62]. Ideally a TF distribution needs to have a high clarity, zero cross terms and low computational complexity.

A variation of the conventional Fourier transform in the joint time–frequency domain is the STFT which is used for computing the frequency and phase content of a localized signal over time. STFT extracts multiple frames of the signal to be analyzed using a moving-time window. The windows width is kept so narrow that the frame is perceptibly stationary for signal analysis [59,64,49]. Depending on the windowing function $h(t)$ we can design the STFT to be a narrowband or wideband transform. Narrow windows do

Table 2

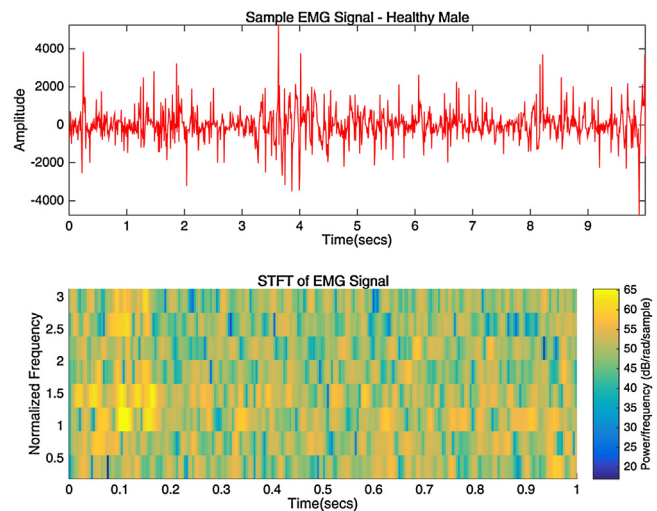
Summary of frequency-domain feature extraction methods.

Method	Advantages	Disadvantages	Applications
DFT/FFT	Easy to implement on hardware Generates sparse or significant features Applicable to finite windowed signals	Cannot be applied to multi-channel signals Transforms only selected signal samples/windows Does not work with non-stationary signals Cannot capture instantaneous frequency content	ECG Analysis Audio spectral analysis EEG filtering Signal separation and compression
DCT	Cosine transform only on real signal values Reduce signal into small number of cosine components Energy conservation/compaction Orthogonal transformation	Output is always real-valued, so quantization is needed to get interger-valued output Specify if the cosine function to be applied is odd or even	Audio and image compression Pathological voice analysis
Spectral Estimation	Helps assess frequency and energy content in a signal Find periodicity in short signal window Robust to noise and quantization effects	Cannot estimate densities of instantaneous frequency components Only applicable to finite/stationary signal windows Provide an averaging effect on signal, which might cause information loss	Speaker recognition ECG analysis Image watermarking Audio fingerprinting
Hilbert transform	Preserves signal energy Extracts instantaneous features Define relationship between real and imaginary parts of complex signals	May need additional operations before implementation on discrete signals Helps in noise removal by orthogonal transformation of signal components	ECG, EEG Analysis Auditory prostheses Filter design

not offer us good localization in the frequency domain, but rather in the time domain. When $h(t)$ is infinitely long, STFT turns into Fourier transform, giving us excellent frequency localization, but does not yield any time information. On the other hand when $h(t)$ is infinitely short, we get good time localization. One must also note that the STFT is not strictly a TF representation since the underlying kernel function may change over time. STFT provides us a local analysis framework for non-stationary signal analysis [49]. Based on Heisenberg's uncertainty principle, we cannot find the exact TF representation for a non-stationary signal, but we can only know which frequency intervals are present in which time intervals [65].

The fact that STFT uses a fixed width window makes it difficult to capture all the non-stationary characteristics from a signal due to the time and frequency resolution trade-offs. STFT has been extensively applied in audio signal processing [59,64,49], especially music genre classification [66,67] and speech signal synthesis [64,59,29]. STFT is desirable in applying to uni-modal, uni-variate signals wherein multiple component complexities do not exist and moreover the signal artifacts and noise are very low. STFT features might need a threshold level during extraction and fitting, as the fixed window limits the amount of non-stationary characteristics to be extracted [59]. To illustrate the implementation of STFT, we have taken into consideration a non-linear, non-stationary, multi-potential EMG signal taken from a healthy male subject [56], as shown in Fig. 7.

In order to overcome the limitations of STFT, the wavelet transform was developed to improve TF representation and resolution. The Wavelet transform is obtained in a similar fashion as STFT, i.e. the input signal is multiplied by a function and the transform is calculated separately for different frames of the time-domain signal. Wavelet transform computation is unique in three aspects: (i) Complex wavelet transform could be applied on negative scales, thus taking into consideration the negative frequencies in the Fourier transform, (ii) the window width is varied for every single spectral component for optimized resolution results and (iii) the signal is decomposed into wavelets. Scaling can cause either dilation or compression of signal, thus generating opportunities for multiple applications of the wavelet transform [68–72,64]. Dilated

**Fig. 7.** (A) Sample EMG signal, (B) STFT/spectrogram.

versions could be cascaded in order to develop newer concepts such as scattering transforms or some similar deep belief networks for signal decomposition; whereas compression versions could be used in compressive sensing technologies. Unlike the STFT which exhibits constant resolution at all times and frequencies, the wavelet transform has a good frequency and poor time resolution at low frequencies and vice versa [68–70].

It must be noted here that the wavelet transform approach is to simultaneously have global and local views of the signal – thus giving rise to the theory of Multi-Resolution Analysis (MRA) which operates on two criteria: (i) the scaling function must be orthogonal to its integer version, and (ii) the subspaces covered by the scaling function at low scales must be nested within those covered at high scales. If we choose the best wavelet adapted to our input signal, or we truncate the coefficients below a pre-defined threshold, we can then sparsely represent the data. This property makes wavelets an excellent tool for data compression; for example, security agencies

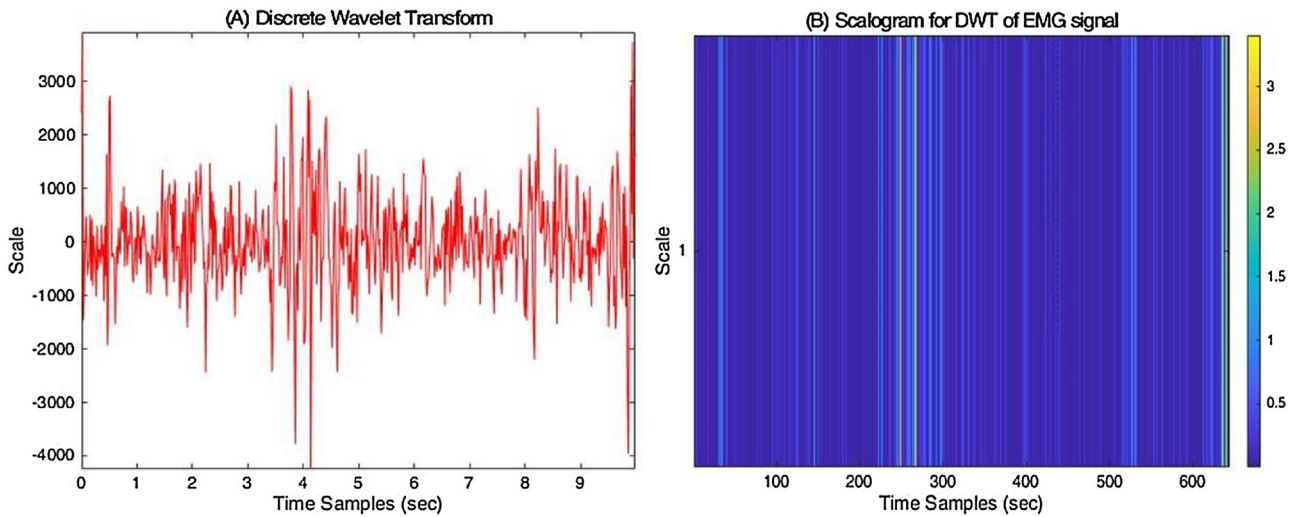


Fig. 8. (A) DWT, (B) scalogram of DWT.

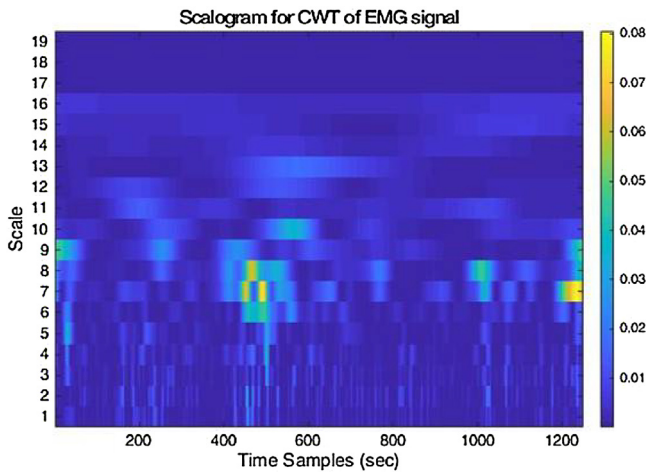


Fig. 9. Scalogram of CWT.

uses wavelet coding in fingerprinting applications [73]. Wavelet transform has been extensively applied in audio signal processing [74,72,64], magnetic resonance imaging (MRI) and computer vision [75], EEG analysis [76] [77], ECG analysis [78], medical image classification [79,68,72]. Fig. 8 illustrates the implementation of DWT on the sample EMG signal (Fig. 7A).

The wavelet transform has two main variations depending on the use of orthogonal and non-orthogonal wavelets as basis functions. The Discrete Wavelet Transform (DWT) decomposes the signal into a set of functions which are orthogonal to its translation and scaling; thereby highlighting as to why it is extensively used in denoising, signal processing and data compression. The Continuous Wavelet Transform (CWT) on the other hand returns an output vector which is one dimension larger than the input signal data. For 1D data we get a TF representation, which implies that the CWT is essentially just wavelet transform. CWT employs non-orthogonal wavelets as basis functions which renders the output vector values to be highly correlated [80–82,69,83]. This improves the signal visualization in higher dimensions, but is not effective enough from a signal classification perspective. We can apply the CWT to discrete time series as well, in the form of discrete time continuous wavelet transform (DT-CWT). CWT has been widely used in cardiac signal processing such as ventricular fibrillation analysis [80–82]. Fig. 9 shows how CWT works on a non-stationary EMG signal (Fig. 7A).

Similar to the wavelet transform, the Wavelet Packet Transform (WPT) attempts to extract transient signal information from a non-stationary signal in the TF plane. WPT is one of the few methods that is highly adaptable to signal characteristics in order to perform effective feature construction. It could be considered as a natural extension of the wavelet transform, wherein the user is able to view a level-wise transformation to TF domain. In this technique, we iteratively split the signal into an approximation and detail [64,84,19]. The top level of the wavelet packet decomposition is a time domain representation of the input signal. As we go down each level of decomposition, we observe that there is a decrease in temporal resolution and a corresponding increase in the frequency resolution, which also leads to lower noise interference. It is a recursive process of wavelet filter-decimation operations wherein we can assume each stage to be a STFT with varying window sizes. In most applications, WPT has been employed to extract features such as energy content, entropy and sub-band correlations [84,19]. Being a tree-based feature extraction technique, it is evident that feature set creation and feature selection is fairly an easy task to accomplish if we apply a simple k -means classifier at each stage in order to look for robust features. Additionally one can observe that since WPT is able to retain approximate and detail features, it is easy to de-noise and reconstruct the signal without significant information loss. WPT has found its applications in non-stationary biomedical signal processing such as knee sounds analysis [62,29,85] and hearing aids development (audio signal processing) [29].

A variant of the TF representation is the ambiguity domain (AD) which provides better spatial domains for data transformation and signal processing with regards to target applications. The AD is represented using an ambiguity function (AF) which is a TF correlation function, and calculates the total energy and Doppler frequency of a non-stationary signal. For a given time–frequency distribution we compute its ambiguity domain properties by taking the Fourier transform of the signals Wigner-Ville distribution [86–88]. Based on the literature [89,83,90–92] the ambiguity domain has four distinct characteristics: (i) cross-terms, which falsely indicate the presence of signal components, and are mostly used in source separation and phase estimation, (ii) auto-terms – signal terms/values corresponding to each time–frequency pairs; (iii) the ambiguity kernel function could be used to obtain the signals equivalent TF structure, and (iv) the structure of spread is highly dependent on the amount of coherent components in the signal.

Similar to TFD property, the AF allows for obtaining the time–frequency spectrum from the signal, which further helps

in extracting instantaneous and delay features from the non-stationary signal. Post segmentation we use the ambiguity function to map the signal to the ambiguity domain which yields us a multivariate signal. De-noising is done using soft thresholding by filtering out cross terms using a low pass filter. Owing to its specific characteristics of ease of representation of cross terms, and retaining auto terms, the AF is easy to understand and implement on non-stationary signals; but it also requires a robust classifier for support [6]. The AD has similar benefits like TFD, such as localization of spectral components for extracting instantaneous frequency, and group delay features for stochastic processes/signals [6].

Following up on the ambiguity domain, we touch base with the Ramanujan Fourier Transform (RFT) [91,83] which transforms a one dimensional non-stationary signal into a 2D TF plane with robust feature map. The Ramanujan Sums (RS) are exponential sums with their exponent defined over irreducible fractions. Their orthogonal property (which is good for energy conservation) coupled with infinite impulse response makes them good alternatives to Fourier transform. They are used for extracting independent quasi-periodic coefficients from non-stationary signals, which are comparatively less in number than Fourier coefficients. The inherent orthogonal property of these sums yields us convergent functions which is highly suited to signal processing applications. Since the RFT decomposes a signal into co-prime resonances, we only need a few samples for estimation, thus indicating sparsity property for periodic-like signals. Also we need to calculate the Ramanujan basis function only once, over which the entire signal can be projected. Being able to process low frequency signals, compared to Fourier transform, RS is more resilient to noise. When applying RS for sparse signal processing, we need to ensure the sparsity of the coefficients in order to check possibility for signal discrimination or reconstruction. Typical applications of Ambiguity domain include non-stationary biomedical signal analysis such as Gait [93], ECG, EEG analysis [89,83,90–92,88,94].

Decomposing a signal by transforming it into a 2D matrix in the TF plane, unravels hidden information and robust features. The Non-Negative Matrix Factorization (NMF) is one such matrix decomposition technique which is used for quantifying the time–frequency distribution of a non-stationary signal. Unlike the conventional methods, the Adaptive TF Decomposition allows for handling non-stationarity of a signal without segmenting it into short intervals, thereby allowing the temporal and spectral localization of signal components. Once the signal has been transformed to the TF plane, we assume its TFD to be a matrix and decompose it using NMF. Features are extracted using NMF over a positive, cross-term free and high resolution adaptive TFD which is selected using a matching pursuit technique. Using an initial estimate of the basis and encoding matrices, the NMF technique aims towards minimizing a given cost function. Apart from the traditional TFD features such as coefficient vectors, spectral moments, base vectors, sharpness and derivative sums, the NMF also generates an important feature – sparsity [87,74,86].

The sparsity of the feature vectors helps in measuring and approximating the abnormality present in the signal by distinguishing between the transient components and the natural components. Current challenges to NMF include application to multi-variate and multi-channel signals wherein resolution to higher dimensions is preferred [95,47,88,96,66]. NMF has a major drawback that the extracted features have a very high dimension since the length of each vector is proportional to the sampling frequency. Most variations of NMF are subjected to sparsity and spectral localization constraints [97]. Despite this, NMF is one of the key TF matrix decomposition methods and has a multitude of applications in biomedical such as EMG analysis [98,99], EEG analysis [96,100], and audio scene classification [95,101]. Figs. 10 and 11

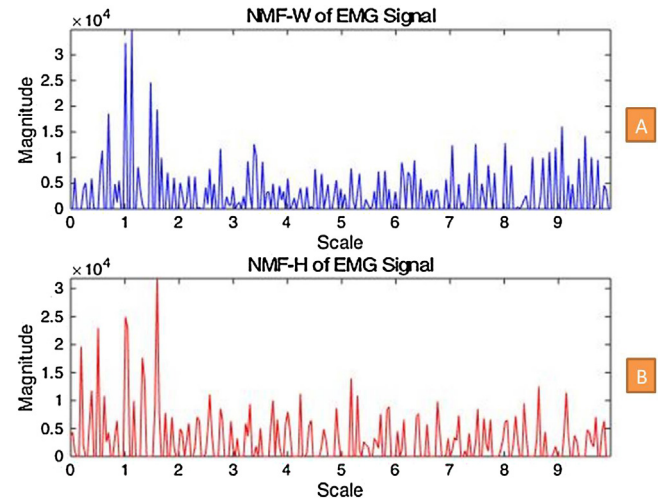


Fig. 10. (A) Non-negative factor W, (B) non-negative factor H.

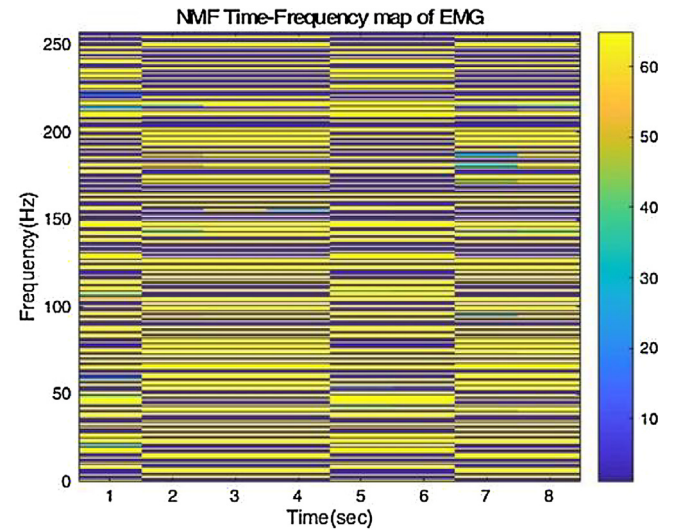


Fig. 11. Time–frequency map for NMF of the EMG signal.

illustrate the application of NMF and its dimension reduction capability, on a sample EMG signal [56].

As many physiological signals are modulated or impacted by circulatory, respiratory and autonomic nervous system (ANS), the signals exhibit dynamics that make them non-linear, non-stationary and non-Gaussian. Most feature extraction methods work assume the signals to be linear, stationary and Gaussian, and therefore extracted features provide only an approximate representation of the true physiological characteristics of the signal. Techniques such as wavelets and time–frequency have addressed the non-stationary aspect to some extent, but many features extracted from wavelet and time–frequency domain fail to characterize the true non-stationarity as they are averaged in time domain, and thereby losing the temporal dynamics of the signals.

For a given signal $x(t)$, if its TFD is given by the relation $W(t, \omega)$, then the total energy of the signal is given as [102],

$$\int \int W(t, \omega) = \int |x(t)|^2 dt = \int |X(\omega)|^2 d\omega \quad (1)$$

where $X(\omega)$ is the Fourier transform of $x(t)$. This indicates that at a particular t and ω , $W(t, \omega)$ gives the fractional energy of the signal. The TFD could be considered as a two dimensional probability density function with t and ω as random variables. For the TFD to be

a valid probability density function (PDF), it must satisfy the positivity criterion, i.e. $W(t, \omega) > 0 \forall t$ and ω . Negative TFDs cannot be extended as PDFs, and may cause interpretation problems when performing objective feature extraction and pattern identification. Integration along frequency axis yields the instantaneous energy of the signal, whereas integration along the time axis yields the power spectral density of the signal [102].

$$\int W(t, \omega) d\omega = |x(t)|^2 \quad (2)$$

$$\int W(t, \omega) dt = |X(\omega)|^2 \quad (3)$$

Few feature extraction techniques that treat the TFDs as probability density functions (PDFs) and compute marginal probabilities to provide marginal densities, and ensemble averages to give first and second order moments are capable of capturing time-varying frequency and time-varying bandwidth instantly also known as instantaneous frequency and instantaneous bandwidth. Instantaneous frequency and instantaneous bandwidth provide true time-varying spectral features which could be tracked in monitoring highly time-varying processes such as heart-rate variability, tracking arrhythmic activities such as ventricular fibrillation related spectral changes [102]. Local expectation values can help evaluate and track non-stationary features such as instantaneous frequency or group delay. They can be computed by applying the expectation operator either along the time or frequency axis. The instantaneous mean frequency, given by the time-varying first moment of the TFD along the frequency axis is given as,

$$E_t \omega = \frac{1}{|x(t)|^2} \int \omega W(t, \omega) d\omega \quad (4)$$

And the group delay along the time axis is expressed as,

$$E_\omega t = \frac{1}{|X(\omega)|^2} \int t W(t, \omega) dt \quad (5)$$

Similarly, the instantaneous bandwidth can be expressed as [103],

$$\sigma_\omega = \left(\int (\omega - \langle \omega \rangle)^2 |X(\omega)|^2 d\omega \right)^{1/2} \quad (6)$$

where $\langle \omega \rangle = \int \omega |X(\omega)|^2 d\omega$

In certain cases, the TFDs which are treated as PDFs provide a measure of uncertainty, i.e., in case of energy distributions appearing in high probable time and frequency concentrations, the uncertainty is low and vice-versa. Such interpretations lend itself well in extracting information theory metrics based on entropy. The entropy could be treated as simple non-linear features as the computation of it involves non-linear quantification via logarithmic expressions. The Shannon entropy measure can be computed within non-overlapping (upper u_B and lower l_B) frames $l_B, u_B \in B$, where B is the bandwidth of the selected signal frame, as,

$$SE_B = \sum_{\omega=l_B}^{u_B} |X(\omega)| \log_2 |X(\omega)| \quad (7)$$

Similarly, we can also measure the spectral distribution of the signal using Renyi entropy, given by the relation,

$$RE_B = \frac{1}{(1-k)} \log \left(\sum_{\omega=l_B}^{u_B} |X(\omega)|^k \right) \quad (8)$$

It could also be seen that such a logarithmic-based features might provide an inherent deconvolution property of convolved or multiplicative sources by taking advantage of the Fourier property that – multiplication is convolution in the transform domain, and multiplication becomes additive due to logarithmic transformation. Such logarithmic-based features in the form of Mel-frequency Cepstral Coefficients have shown good performances in characterizing pathological voice signals as dominant features.

It is also possible that when certain activities such as linearly frequency modulated signatures such as chirp get represented as a straight line pattern in a time–frequency plane, then they become an easy pattern to detect using simple line detection techniques. This could be achieved by treating the time–frequency distributions as images with pixels denoting normalized energy values, and time and frequency representing rows and columns of images. Efficient straight line detection algorithms such as the one based on Hough transform can be implemented such that the line detection becomes a linear FM signature detection algorithm.

The Hough transform is used for computing the number of features or image points (in pixels or feature values), which satisfy a parametric constraint [104,105], $f(W, \Omega) = 0$. Here, $W = (w_1, w_2, \dots, w_N)^T$ is a point in the TFD, $W(t, \omega)$ of the signal, and $\Omega = (\Omega_1, \Omega_2, \dots, \Omega_M)^T$ is a point in space of parameters, known as the Hough space. Depending on each feature point, we can use the constraint to interpret detection of straight line, curve or a surface. Thus, we can map the constraint into Hough space by evaluating, $W: f(W, \Omega) = 0$. Similarly other patterns that organize well within the two dimensional time–frequency plane that could be mathematically expressed lends itself well in implementing complex feature extraction approaches [105]. This approach has been found useful in characterizing biological sounds such as the one emitted by bats and dolphins. In the biomedical context, TF pattern recognition from images has been applied to detect chirp like patterns emitted from knee [104,105].

Evolutionary methods such as the STFT have given us a solid platform for tackling non-stationarity in signals by using a windowing approach. Although STFT performs very well by extracting low level features in the time–frequency plane, the transformation from 1D to 2D for signal analysis becomes computationally intensive for complex signals. Despite its local-framework generation, the limited time–resolution, and redundant feature generation has led to development of newer techniques such as wavelet transform, wavelet packet transform, and NMF. Modern day biomedical signals are multi-dimensional, multi-variate and multi-component in nature, and hence their transformation to 2D TF plane using robust methods is imperative. **Methods such as wavelet transform exploit the non-stationary signal by extracting hidden information using a varying window function, which yields us transients from the sharp discontinuities in the signal.** The number of features generated is comparatively lower than STFT and sparsify the complex signal in 2D TF plane.

A better option than wavelet transform is its deep belief successor Wavelet Packet transform which performs iterative analysis on the signal in a cascading fashion by employing the same mother wavelet (or window-function) at each stage and extracting instantaneous information/features. Non-negative matrix factorization is also an excellent method for non-stationary signal analysis, which matrices a one dimensional signal into two dimensions through a TF transformation. The matching pursuit algorithm matches random signal patterns using a dictionary and transforms it to a 2D TFD structure. This two-dimensional structure is then processed using NMF, thus yielding us robust sparse features which minimize the complex signal into minimum representation components. It must be noted here that NMF has been used in conjunction with sparse signal processing techniques for signal classification. **Table 3 highlights key aspects of time–frequency methods.**

Table 3
Summary of joint time–frequency domain feature extraction methods.

Method	Advantages	Disadvantages	Applications
STFT	Handles non-stationarity using windows Localized TF representation for each windowed signal segment Can be narrowband or wideband, depending on application or signal properties	Limited time/frequency localization Fixed window size cannot capture all non-stationary characteristics Redundant feature generation Applicable mostly to uni-modal, uni-variate signals	Speech analysis Music classification EMG analysis
CWT	Considers negative frequencies in Fourier domain Varying window width captures non-stationary characteristics Varying TF resolution Small number of components called Wavelets Possibility of signal dilation and compression	Highly correlated output vector values lower signal classification rate Computationally intensive for complex signals	Ventricular fibrillation EMG, EEG, ECG analysis Medical Image classification Magnetic Resonance Imaging
DWT	Decompose signal into orthogonal functions Extensively used in denoising, data compression	Applicable only to discrete time-series Computationally intensive for complex signals Highly correlated output coefficients lower signal classification rate	Signal compression Magnetic Resonance Imaging ECG, EMG analysis Fingerprinting applications
WPT	Improvisation of Wavelet transform Level-wise decomposition into features Signal dimension reduction and better feature space visualization Lower noise interference Can extract instantaneous features Easy to denoise and reconstruct the signal	Computationally intensive due to TF transform at each stage Needs large amount of data for better classification Cannot include signal structure and morphology information during feature extraction	Knee sound analysis Hearing aids development Magnetic Resonance Imaging
Ambiguity FunctionRFT	Better spatial domain transformation compared to TFD Improved signal visualization in ambiguity domain Filters cross terms and retains auto terms Improves signal denoising Orthogonal RFT conserves energy RFT is good alternative to Fourier transform RFT generates sparse features	Highly dependent on coherent signal components Signal itself is a window function, which increases computational complexity	Gait analysis EEG, ECG analysis
NMF	Quantifies TFD for a non-stationary signal Sparse feature generation NMF could double up as machine learning tool Positive, cross-term free features	Needs initial estimate of basis and encoding matrices Not easily applicable to multi-variate, multi-channel signals Extracted features have higher dimensions Most NMF variations have sparsity and spectral localization constraints	EEG, EMG analysis Audio scene classification

5. Decomposition and sparse domain feature extraction

5.1. Signal decomposition domain feature extraction

In continuation from previous section which emphasizes on sparse signal processing methods, in this section we describe various algorithms and techniques which decompose a non-linear, non-stationary input signal into meaningful components or features which could be construed as sparse in nature. The decomposition methods discussed here extract features from a time–frequency–energy (TFE) analysis perspective and have evolved based on three important premises [106]: (i) balancing the trade-off between accurate temporal information and frequency information due to Heisenbergs uncertainty principle. The temporal information could be calculated using the time-scales

determined by the frequencies present in the signal; (ii) the decomposition should be basis free, i.e. we should avoid considering a priori information about the signals content or composition or nature during analysis; and, (iii) attempt to perform efficient signal analysis in real-time.

Analyzing hidden information within multi-component signals in order to extract underlying meaningful patterns requires us to breakdown the signal “microscopically”. Decomposition in general means to breaking down any complex process into individual constituent components. Applying Empirical Mode Decomposition (EMD) is a novel approach for feature extraction and pattern classification, wherein we break the complex signal into Intrinsic Mode Functions (IMFs). These IMFs are artificially generated by the EMD process, and may not be part of the original signal, but they are highly useful in decrypting patterns from the signal. The EMD was

proposed as an integral part of the Hilbert-Huang Transform (HHT) [106–109] which operates as follows: (i) using the EMD algorithm, we compute the IMFs; (ii) calculate the instantaneous frequency spectrum using Hilbert transform.

EMD is a user-controlled iterative decomposition process, and results in a group of frequency ordered components. Successive ordered IMFs have lower variations in amplitude and frequency. IMF extraction (or sifting) involves the following two steps [107]: (i) identify all the extrema in the signal, and (ii) produce the lower and upper envelopes by connecting local maxima and minima using a cubic spline line. De facto standards indicate that sifting stops when the residue signal does not contain more than two extrema. This sifting process has two goals: (a) separating high-frequency, small-amplitude components, superimposed on components with large amplitude and low frequency, and (b) smoothing uneven amplitudes in the IMF. But these goals conflict non-stationarity of the signal since riding components might be transient in nature and may vary drastically in amplitude. Also when we try to smooth out uneven amplitudes using sifting, it could prevent precise extraction of riding components. Note that repetitive sifting can cause spreading of TFE information across different decomposition levels which would not exhibit the intrinsic characteristics of the signal. Fig. 12 illustrates how EMD would be well suited for a non-linear, non-stationary EEG signal, and help in extracting significant features through IMFs.

EMD is useful in analyzing non-stationary and non-linear signals, and has been successfully applied in applications such as EEG analysis [110,111], pathological voice signal analysis [107,112–114]. Being a data-driven technique, EMD is highly adaptive and performs exceptionally well when the signal tests positive for sparsity. Also because of the shorter spectrum from the Hilbert transform, feature extraction could be possible in 1D instead of the usual 2D. Decomposing a complex signal iteratively into smaller IMFs, which provide good representation of the signal in smaller dimensions and allow for reconstruction, suffices that EMD is a highly efficient sparse domain signal processing technique. In addition to using EMD for feature extraction applications, it also helps in signal processing by performing pre-processing tasks such as de-noising and de-trending of signals.

One of the most recent signal decomposition methods overcoming the limitations of conventional methods such as Fourier transform, or wavelet transform, and EMD is the Intrinsic Time-scale decomposition (ITD) introduced by Frei and Osorio [106] in 2007. ITD has the capability of performing TFE analysis of non-linear, non-stationary signals, for extracting meaningful features which sparsify the input data. In simple terms, ITD is a two-step operation [106]: (i) decomposition of signal into a sum of Proper Rotation Components (PRC) for the instantaneous parameters (frequency and amplitude) are well defined; and, (ii) decomposition of signal into a monotonic trend thus preserving the signal order. The ITD retains accurate temporal information about critical events in the signal, such that the temporal resolution is equal to the time-scale of extrema occurrence in the input signal [106]. ITD overcomes the laborious tasks of sifting (refer to EMD) and splining of non-stationary signals which saves a lot of computational overhead.

Using ITD we can develop a real-time signal filter which can extract precise temporal information with an accurate temporal resolution of the extrema in the time-scale plane. ITD filters can extract features from non-stationary signals at their naturally occurring time-scales, thus preserving their morphology and relative measures [106]. After decomposing the signal into PRC and a residual signal, we can further analyze the rotations to compute instantaneous amplitude, phase and frequency information [106]. Additionally it should be noted that ITD works in an iterative fashion, i.e. after the first stage of decomposition, ITD could

also be applied from the baseline signal for extracting low level features. This adds up in developing a robust feature set from the non-stationary signal for pattern classification problem. Instead of Hilbert transform, we apply piecewise wave-based approaches to compute instantaneous TFE values [6,106]. ITD is ideally suitable for analyzing non-stationary biomedical signals such as EEG signals especially in epilepsy detection. ITD could also be considered to be a signal sparsifying technique [106].

Similar to ITD, signal dimension reduction and feature extraction could also be achieved using time-frequency decomposition techniques, by applying orthogonal basis or kernel functions to non-stationary signals, thereby reducing them to a smaller subset of meaningful features. This approach also ensures energy conservation and prevents information loss. In case of signals which do not decompose easily using orthogonal transformations, we can employ the use of greedy search algorithms, which decompose a non-stationary signal by matching it using a dictionary of basis functions. These basis functions perform one-on-one matching with each signal component in order to generate a compact TF representation.

One such method of basis function application is the matching pursuit (MP) algorithm for TF decomposition. This algorithm transforms a given signal by decomposing it using basis functions or matching vectors selected from a set of functions called dictionary [102]. Therefore, a non-stationary signal $x(t)$ is projected onto a subset dictionary of matching functions obtained by scaling, translating and modulating a window function $h(t)$, such that,

$$x(t) = \sum_{n=0}^{\infty} a_n h_{\gamma_n}(t) \quad (9)$$

where h_{γ_n} is,

$$h_{\gamma_n}(t) = \frac{1}{\sqrt{s_n}} h\left(\frac{t - p_n}{s_n}\right) \exp([j(2\pi f_n t + \phi_n)]) \quad (10)$$

and a_n are the expansion coefficients. The scaling parameter s_n controls the width of the window function $h(t)$, whereas p_n controls the temporal placement. The parameters f_n and ϕ_n depict the frequency and phase information of the exponential function in Eq. (10). The norm of $h_{\gamma_n}(t)$ is restricted to 1 using the factor $\frac{1}{\sqrt{s_n}}$; and γ_n represents the set of all parameters as (s_n, p_n, f_n, ϕ_n) .

The MP-TFD algorithm works as follows [17]

- The signal $x(t)$ is projected onto the dictionary of atoms, and the first projection decomposes the signal into two parts as,

$$x(t) = \langle x, h_{\gamma_0} \rangle h_{\gamma_0}(t) + R^1 x(t), \quad (11)$$

where, $\langle x, h_{\gamma_0} \rangle$ denotes the inner product or projection of the signal $x(t)$ with the first atom from the dictionary, which is $h_{\gamma_0}(t)$. The term $R^1 x(t)$ is the signal's residue left after approximating it in the direction of $h_{\gamma_0}(t)$.

- The process in Eq. (11) is iterated by projecting the residue on subsequent basis functions in the dictionary for M iterations, given by the expression,

$$x(t) = \sum_{n=0}^{M-1} \langle R^n x, h_{\gamma_0} \rangle h_{\gamma_n}(t) + R^M x(t), \quad (12)$$

with $R^0 x(t) = x(t)$. In order to set a limit M , we can choose from the following two approaches:

- Using a predefined number of M iterations, or,
- Checking the energy of the residue $R^M x(t)$, i.e. A high value of M and zero residue, can decompose a signal completely but it would significantly increase the computational complexity of the algorithm.

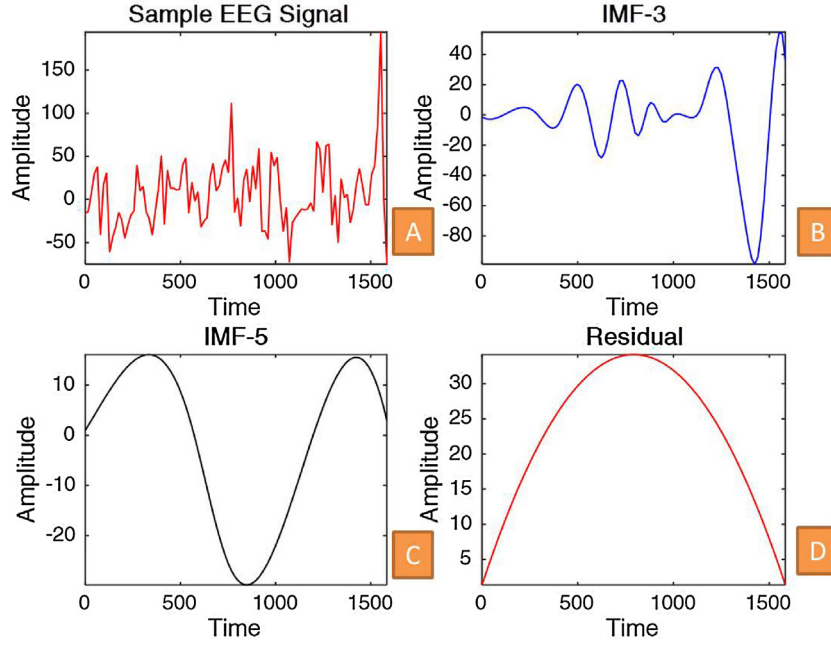


Fig. 12. (A) Sample EEG signal, (B) IMF-3, (C) IMF-5, (D) residual.

We can find the first M coherent structures using a signal decay parameter $\lambda(m)$, given by the expression,

$$\lambda(m) = \left(1 - \frac{\|R^m x\|^2}{\|R^{m-1} x\|^2} \right)^{1/2} \quad (13)$$

Here $\|R^m x\|^2$ is the signal energy level at the m th iteration. The signal decomposition is iterated till the decay parameter stops reducing, the stage at which the selected components represent coherent structures and the residue components represent incoherent structures in the signal with respect to the dictionary. The residue can be considered as noise and ignored, since it does not indicate any localization of the signal. From the M coherent structures, we can then reconstruct the signal as,

$$x(t) = \sum_{n=0}^{M-1} \langle R^n x, h_{\gamma_n} \rangle h_{\gamma_n}(t), \quad (14)$$

which also denotes the denoised version of the original signal.

A TF representation for signal decomposition can be obtained by taking the Wigner-Ville Distribution (WVD) of the dictionary atoms using the expression [17],

$$W(t, \omega) = \sum_{n=0}^{M-1} |\langle R^n x, h_{\gamma_n} \rangle|^2 W_{\gamma_n}(t, \omega) + \sum_{n=0}^{M-1} \sum_{m=0, m \neq n}^{M-1} \langle R^n x, h_{\gamma_n} \rangle \langle R^m x, h_{\gamma_m} \rangle W_{[\gamma_n, \gamma_m]}(t, \omega) \quad (15)$$

The double sum in Eq. (15) denotes the cross terms of the WVD. Ignoring or rejecting these, we get the resulting TF representation as,

$$W'(t, \omega) = \sum_{n=0}^{M-1} |\langle R^n x, h_{\gamma_n} \rangle|^2 W_{\gamma_n}(t, \omega) \quad (16)$$

Eq. (16) is known as matching pursuit TFD (MPTFD), and is suitable for non-stationary and multicomponent signal analysis, and in case of signals with unknown SNR (Signal to Noise Ratio). Although

MPTFD satisfies energy and hence information conservation, it does not meet the time and frequency marginal requirements. In order to resolve this, the algorithm needs to follow a Minimum Cross-entropy optimization procedure [17], for removal of cross terms. The matching pursuit algorithm has been extensively applied in audio signal processing and knee-sound vibration analysis [102,17].

Signal processing applications deal with numerous issues in analyzing matrix data with real or complex values. Breaking or decomposing this matrix data into meaningful pieces requires the application of singular value decomposition (SVD) [115,74,116]. Similar to NMF, the SVD technique tends to decompose a large matrix into smaller number of meaningful features which would be useful in pattern classification. In other words, SVD could be thought of as a factorization method. The mathematics behind SVD originates from the theorem that – any rectangular matrix X of size mn composed of real or complex values can be expressed as a product of 3 matrices: an orthogonal matrix O , a diagonal matrix D , and the transpose of an orthogonal matrix V . SVD takes a high dimensional dataset, and reduces it to a lower dimensional space which uncovers a composition of the original data, and rearranges it in a descending order of most variation to the least. SVD allows for threshold levels to be set, in order to ignore data variations. This greatly helps in reducing data size. This being said, we can firmly say that SVD is a Signal Sparsifying Decomposition technique [116]. We speculate that SVD could double up as a feature selection or feature set reduction technique, considering its thresholding and sorting capabilities. Applications of SVD include data compression, noise reduction, and biomedical signal analysis such as ECG signals [117,116] and EEG signals [100]. One must note that although SVD may be good in dimension reduction sparsification, the thresholding of most variable components and hence their filtering, does pose a challenge with respect to signal or information loss. Also it should be considered that even if SVD can solve trivial non-stationary problems, it is still not precise and has poor convergence. For the readers additional information SVD also has a dictionary learning variation known as K-SVD, which is based on K-means clustering algorithm (Fig. 13).

Emerging multi-sensor technologies and the subsequent big data generation has also led to developments in multi-dimensional

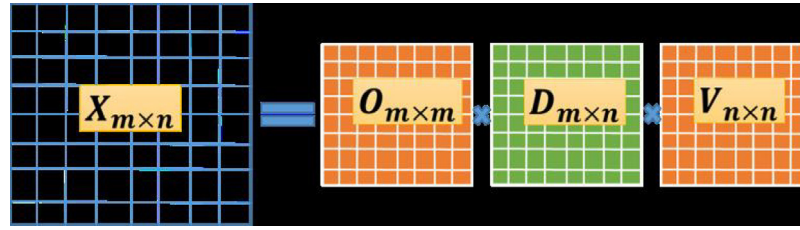


Fig. 13. singular value decomposition (SVD).

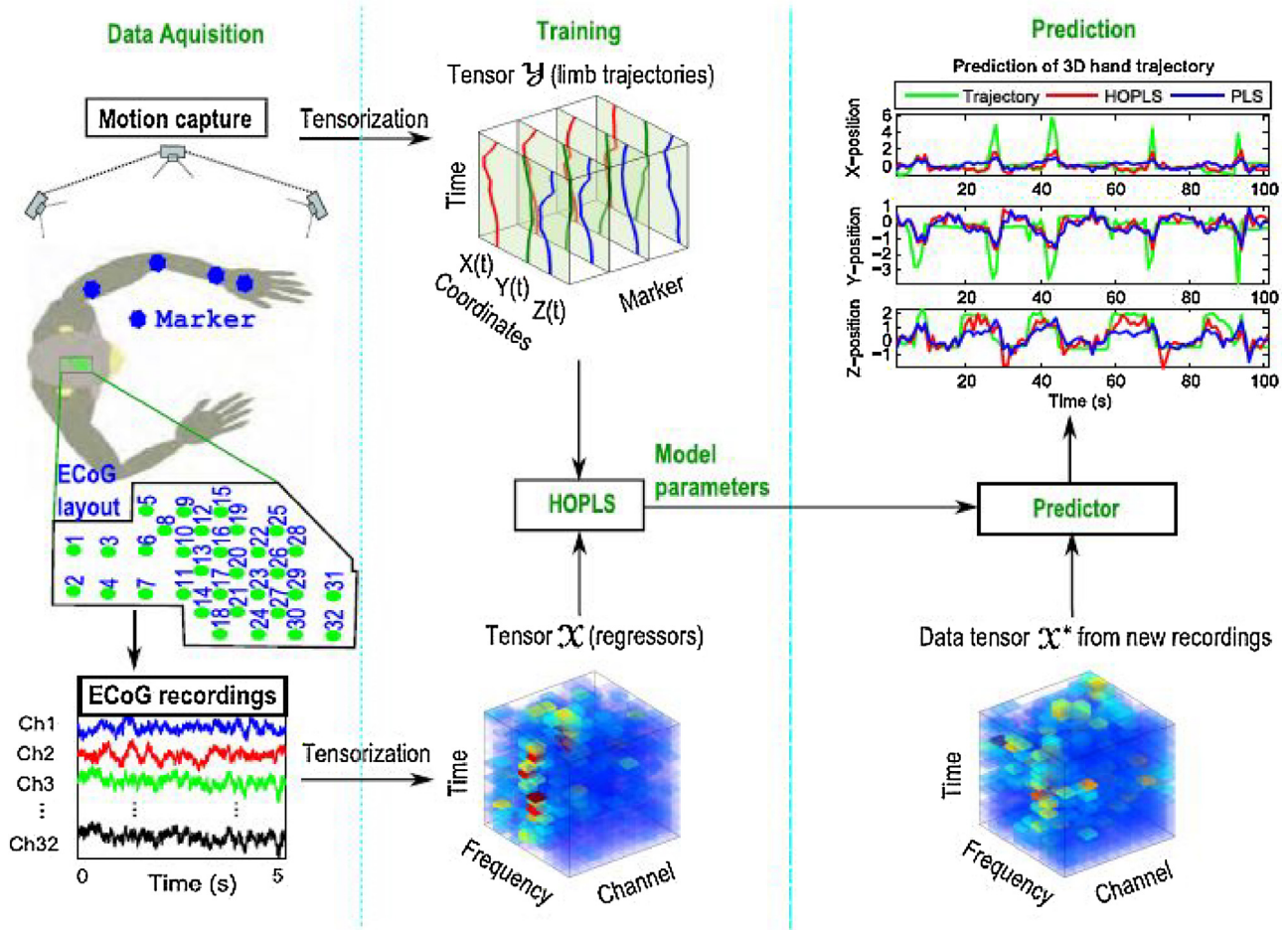


Fig. 14. Example of tensor analysis [118].

signal processing techniques. Conventional signal processing techniques wherein data visualization is done in 1D or 2D (matrix), are not conducive for multi-channel non-stationary datasets, and may need an upgrade in higher dimensions for feature extraction. Tensor Decomposition is one method of decomposing multi-channel, multi-dimensional non-stationary signals into meaningful features for pattern classification. Multiway arrays or Tensors are higher dimensional dataset [118]. One may ask – why do we need to transform simple one dimensional data into a higher space? The simplest answer to this would be that, we do tensorization of signals in order to remove artifacts, and improve signal visualization in order to reveal hidden and latent features. Signal visualization greatly impacts the method we chose for signal analysis, feature extraction and signal classification. Tensor structure is based on multi-linear algebra which greatly helps in adapting to the input signal structure for dimension transformation. A tensor can be imagined as a multi-index numerical array, wherein the tensor order could be defined by

the number of parameters such as time, frequency, space, classes and dictionaries. Fig. 10 illustrates how tensors are created from non-stationary signals (Fig. 14).

Once our signal has been tensorized, it is easy to analyze patterns and extract classes from the signal. While 1D and 2D representations of signals are mostly scalar in nature, tensors are highly vectorized which could provide additional information about the signal emitting source. For example, while doing EEG analysis, tensorizing multichannel EEG data will not only highlight the brain state but also give us insights into the specific brain region or lobe wherein the electrical activity is occurring [118]. The higher dimensionality of tensors gives us the benefits of choosing constraints, generalizing signal components, developing compact representations and of course, the uniqueness in signal decompositions. Feature extraction from tensors involves their decomposition or dimension reduction or Matricization in order to extract meaningful low level features. Prominent decomposition methods include:

Table 4

Summary of decomposition domain feature extraction methods.

Method	Advantages	Disadvantages	Sample applications
EMD	Doesn't need a priori information Useful for decrypting hidden information in a signal User-controlled decomposition Easy to decompose a complex signal for classification and reconstruction Helps in denoising and detrending Adaptive and helps in sparse feature generation	Sifting can ignore transient features Envelope formation could cause TF energy to smear across different levels High computation time for sifting/splining No decision rule to select suitable IMFs for features	EEG Analysis Pathological Voice analysis Time-series analysis
ITD	Captures instantaneous features Doesn't need a priori information Retains accurate temporal information Preserves signal order by decomposing signal into a monotonic trend No sifting process required Preserves signal morphology	Features are sometimes not accurate, since no a priori information has been applied Higher computations for complex signals	EEG, EMG Analysis
MPTFD	Preserves signal energy and information Decomposition denoises the signal using an iterative approach Simple to implement using correct optimization procedure Generates compact TF representation	Computationally complex when converging to an accurate solution Dictionary selection could be tedious Does not meet time and frequency marginals	Audio signal analysis Knee vibration sound analysis
SVD	Factorization based decomposition approach Enhances signal pattern visualization using sparse features Allows signal thresholding to control variations Doubles up as feature selection technique	Thresholding could cause signal information loss Computationally intensive technique Decomposition may not be precise, and has poor convergence	Denoising, Compression ECG analysis EEG analysis
Tensor Analysis	Applicable to multi-channel, multi-dimensional signals Helps in denoising and artifact removal Enhances feature space visualization Multi-linear algebra logic adapts to signal structure Vectorization helps find signal source	Computationally intensive for low level feature extraction Feature mapping may not be accurate for complex signals Unnecessary dimension transformation for artifact removal	EEG analysis Atrial-fibrillation analysis

Block Term Decompositions, Tucker Decompositions and Canonical Polyadic Decompositions [118].

It might sound paradoxical about the entire tensorization and feature extraction process from a signal processing perspective. That is, one might feel that we might be unnecessarily going from lower to higher dimensions in order to extract low level signal features. This being said, one must also note that tensor representations give us a virtually sparsified or a compact signal representation, even though the underlying tensor element count might be large.

Though signal decomposition yields us minimal representation they are different from sparse methods in the sense that these methods decompose a signal into smaller representative elements/functions as opposed to matching random patterns in the signal using a dictionary. Decomposition based signal processing works best when the input signal carries considerable artifacts and noise, and our objective is to capture instantaneous and transient information from the signals. An important aspect of many decomposition based signal processing techniques is that they do not make any assumptions a priori before signal decomposition.

And although these techniques do not yield an accurate feature set due to their signal generalization approach, these methods are still popular in reducing a signal to smaller representation since noise removal is effectively done. Also unlike conventional methods which apply a windowing approach to non-stationary signal analysis, decomposition-based techniques do a level-by-level decomposition, thus retaining the scope for capturing transient

signal components which carry hidden information about signal patterns and class. Table 4 highlights key aspects of all afore discussed signal decomposition techniques.

5.2. Sparse domain feature extraction

5.2.1. Sparse representations and dictionary learning

In today's world, huge amounts of signal data gets captured by electronics sensors, which is then used in numerous applications such as analysis, decision making or simple transmission and reception. This captured data is mainly redundant in two aspects [119,120]: (a) there could be multiple correlated versions of the same information in the signal, and (b) each signal is usually densely sampled. It is surprising to know that the relevant information which causes signal generation in a specific pattern, if extracted efficiently, is usually of much smaller dimensions as compared to the original data. This causal information helps in signal analysis and reconstruction, and discrimination within signal classes. This would also mean that there exists a dimensionality gap between the physical processes and their causal observations, which would imply the difference between their respective representations by the sensor and in the physical space [119,120].

Till date, majority of sensors in consumer electronics employ the conventional method of Nyquist sampling (or Shannon theorem), wherein the physical space signal is sampled at a rate which is at least twice the maximum frequency present in the signal. In case of images, this rate is usually controlled by the temporal or spa-

tial resolution. Although this principle enables uniform sampling, it also promotes large memory and power consumption, coupled with computationally expensive pre-processing and conditioning techniques. To overcome these hurdles, the last decade saw the advent of a new theory which promotes sparse representations of signals in the sensing and acquisition process. In this section, we will discuss the concepts of sparse representations and following compressive sensing theory which would give the reader a brief highlight on currently trending signal compression and feature extraction techniques.

The entire objective of feature extraction techniques is not only to identify signal patterns but also reduce the complexity of the signals by transforming them into features (aka dimensionality reduction) and provide a scope for reproducing the signal using its features. Sparse domain signal processing also allows signal processing engineers to do the latter – reproducing signals and identifying patterns in one go. Sparse representations could be literally construed as a minimal representation of a complex time-varying non-linear signal, which would represent characteristic patterns emanated by the signal and aid in reconstructing it if needed. A signal is said to be sparse if it contains very few non-zero components and rest are zero [121]. Following generic steps would help the reader in understanding sparse signal processing methodology [119,120]:

- Select random signal windows with specific frame size.
- Build a dictionary comprising of basis functions, relevant or matching approximately to the signal structure.
- Using the dictionary, we match the random signal windows to specific functions and provide some feedback to the user about the pattern identified – leading to dimensionality reduction in the signal, thus extracting features from it.
- Simultaneously we could also reconstruct the entire signal by matching functions and approximating signal values with respect to the dictionary.

Dictionary creation and learning is a major aspect of sparse signal processing. An ideal dictionary would be the one that can do 100% accurate reconstruction, which also leads us to developing algorithms targeted towards dictionary learning and avoid over-fitting [119,120,122]. In dictionary learning, the input signal is assumed to be a linear combination of basis functions whose coefficients are assumed to be sparse in nature. It could be considered as a resource database comprising of functions known as atoms (each atom being the smallest element representing a basis function), which would be used in multiplication with the sparse coefficient vector for function matching and reproducing the input signal. The dictionary is said to be over-complete when it spans the entire signal space i.e. every input signal can be represented as a linear combination of atoms in the dictionary. But this might lead to redundancy of signal elements during reconstruction thus increasing computational overhead. In order to find the exact representation we relax the dictionary creation requirement, thereby leaving a scope for minimum reconstruction error bounded by a small energy [120].

The reader must note at this juncture that any dictionary learning process involving discrimination of signal patterns, must be supervised for correct labelling. We can do this by modifying the sparse coding step such that the basis function sparsifies the signal for reconstruction as well as distinguishes it from other signal classes. Thus we also aim to minimize the sparse approximation error [120] and ensure that the features selected have minimal inter-class correlation. Another alternative in dictionary learning could be to create incoherent subspaces which represent data in different classes, along with features with low inter-class correlation (Fig. 15).

An interesting point to note at this juncture would be that the objective of feature extraction is also to subsequently achieve dimensionality reduction for pattern analysis. Dimensionality reduction can be explained from a generative/approximation perspective, as well as from a discriminative perspective. The general idea is to compute a subspace or dictionary which can explain the data with a sparse representation [120]. The reduced subspace can be computed using: generative and discriminative methods. A discriminative method [120] is intended towards connecting the original data space and the reduced dimension subspace, so that the original data can be easily segregated/analyzed. This connection or mapping can be linear or non-linear depending on the complexity of the original data space. The intention is to separate the data from different classes in the low-dimensional subspaces. However the discriminative methods do not engage much in extracting meaningful features from the input data, thus rendering themselves to be vulnerable to noise, missing data and improper testing conditions.

On the other hand, a generative method [120] helps in creating data representations which ease analysis, labelling of data as well as extract meaningful features. The role of a generative dimensionality reduction algorithm is to simplify the signal to meaningful features such that it can be easily characterized in the reduced subspace. Common examples include NMF, Principal Component Analysis (PCA), Independent Component Analysis (ICA) [88,96,87]. The process of finding sparse representations is known as Sparse Coding wherein we find a small number of significant sparse coefficients from the input signal (also known as Maximum Likelihood (ML) dictionary learning).

Sparse representation works best for problems in signal reconstruction such as denoising, coding, and inpainting of images. However, if one needs to achieve signal classification, it is important to note that the sparse representation of a given signal must have more discriminative properties for given classes, as compared to the reconstruction error. Conventional classification methods such as linear discriminant analysis (LDA) work under simple assumptions: normal distribution, and the signal to be classified is noiseless. But when this assumption does not hold, these techniques seem to be invalid and inefficient to signal corruptions. That is when sparse techniques come into play and address the problems of noise removal, outlier removal and recovering missing values from the signal. Thus sparse representations combine the best of conventional reconstructive methods such as PCA, ICA and discriminative methods such as LDA, in order to address robust classification when the input signal is corrupted. Sparse representation/approximation algorithms are extensively being applied in biomedical and multimedia signal processing [117,97,123] such as ECG analysis, EEG visualization and image capture in cameras.

5.2.2. Compressive sensing

Although the idea of sparse representations using dictionaries is promising, there still exist some drawbacks which must be addressed in order to develop robust signal acquisition, classification and reconstruction algorithms [121,119,120]. These inefficiencies have been listed as follows:

- Although we can generate sparse representations using good dictionaries, we still need to acquire the full signal in order to generate its sparse or significant coefficients.
- We need to compute transform coefficients for all the signal samples, even though we use only a very small portion of them and discard others. This only increases the number of unnecessary computations.
- We also encounter additional overhead as we need to encode the locations of the large coefficients with respect to basis functions in the dictionary as well as in the signal.

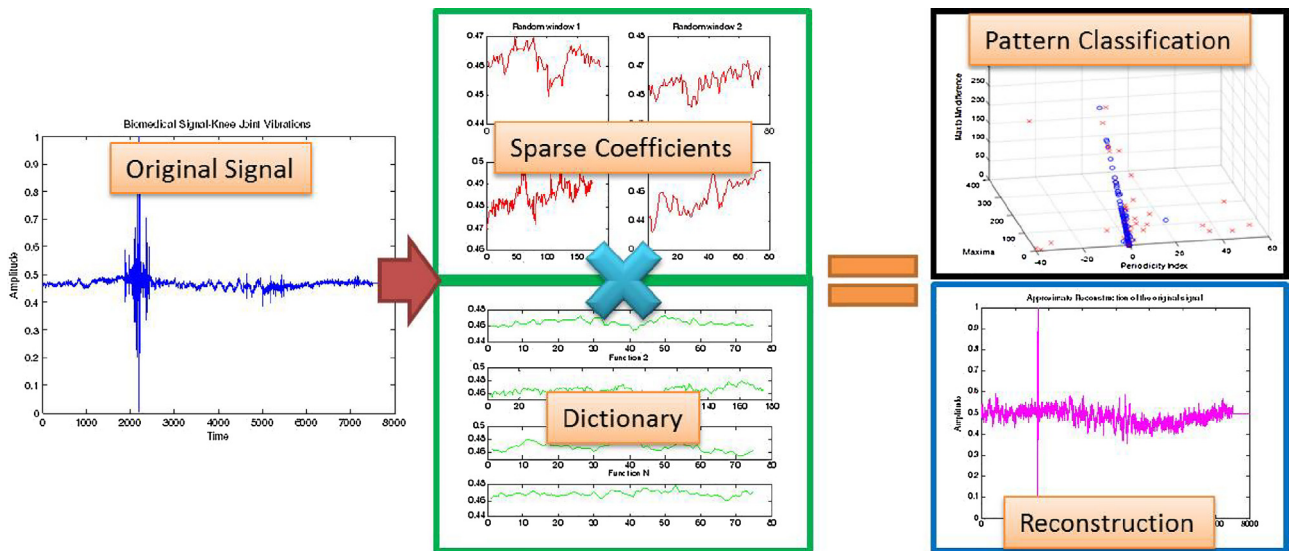


Fig. 15. Sparse representations.

A natural beneficiary to the sparse representation concept is the theory of Compressive Sensing which addresses these inefficiencies by directly generating a compressed signal representation at the source without going through the intermediate stage of acquiring all N samples of the signal. The theory postulates the idea of computing non-adaptive linear projections of a signal that preserve the signal structure. In comparison to conventional Nyquist rate, we can perceive the sampling rate as very high, since only few random measurements are needed for full signal reconstruction using numerical optimization methods. Compressive sensing measurement methods translate analog data into an already compressed digital version, which can then be decompressed into full data using linear programming methods [124]. Being an inherent problem of sparse signal recovery, compressed sensing integrates sparse representations with two highlights of dimensionality reduction: information preserving projections (RIP) and tractable recovery algorithms [121,119,120]. Before we delve into the compressive sensing problem and its proposed solutions, we must briefly understand the two pillars which hold the compressive sensing theory [121,125,124,126,127]: sparsity and incoherence.

- **Sparsity:** suggests that the information content may be much smaller than the length or bandwidth of the signal. A sparse signal has a concise representation when expressed in a proper dictionary. In a sparse structure most samples are close to or zero, and very few have significant values. That is, most coefficients in a signal are small, and the relatively few large coefficients capture most of the signal information. As indicated in previous section, when a signal has sparse expansion, one can discard the small coefficients without much perceptual loss.
- **Incoherence:** indicates that unlike the signal of interest, the sampling or sensing waveforms have an extremely dense representation in the dictionary. For example, a signal may be spread out in the time or frequency domain, but would have a compressed version in the basis domain (Fig. 16).

Compressive sensing is a technique which simultaneously samples and compresses the signal at a reduced rate thereby extracting meaningful patterns from the signal which would be helpful in reconstruction. In definitive terms, Compressed Sensing is a signal reconstruction technique applicable in scenarios wherein the input data is considerably larger for the processing hardware and needs to be reduced or compressed in order to do further analysis of

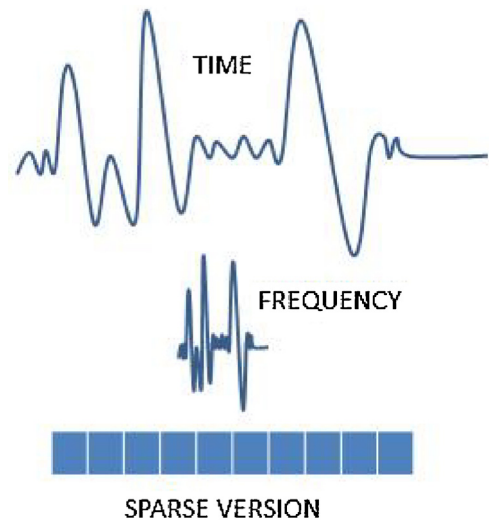


Fig. 16. Incoherence in compressive sensing.

emerging patterns. Though compressed sensing is fairly new area, it is already being used extensively in image processing [128,129], biomedical signal processing such as ECG analysis [125,126,122], and body sensor networks [117]. This technique is most beneficial when signals are sparse, measurements at acquisition end are expensive and measurements at receiver are cheap.

Compressed sensing algorithms can not only recover sparse data, but can also recover compressible data through slight algorithm modifications. These algorithms could also be made robust to noise and signal quantization effects. A major advantage of CS techniques is hardware optimization by reducing computational power during signal acquisition and reconstruction. CS techniques allow the sensors to efficiently capture the information in a sparse signal without compromising information loss from the original signal. Fig. 17 shows a subtle example of how a sample actigraphy signal [35] could be shown that it is sparse in nature, by retaining only its non-zero components.

The core objective of sparse techniques is Minimal Representation with Maximum Information – which results in transforming the large complex signal into small number of representative components in higher dimensions, such that these components contain

Table 5
Summary of sparse domain feature extraction methods.

Method	Advantages	Disadvantages	Sample applications
Sparse Representations	Minimal signal representation Allows classification and reconstruction Sparsity can be achieved using simple or complex transforms Signal analysis is based on dictionary learning Could be generative or discriminative	Mostly applicable to software based signal analysis algorithms Need to acquire full signal for sparsification Need to compute transform coefficients for all samples Additional overhead to encode locations of large coefficients	ECG, EEG, EMG Analysis Image Processing
Compressive Sensing	Minimal signal representation Acquire sparse signal at source Could be implemented on hardware Inherent dictionary learning Allows classification and reconstruction Lower computational power during acquisition and reconstruction	Limited technology to analyse multi-variate, multi-channel signals and systems Not all signals satisfy incoherence property and may need additional transformation	ECG, Gait Analysis Image Processing Body sensor networks

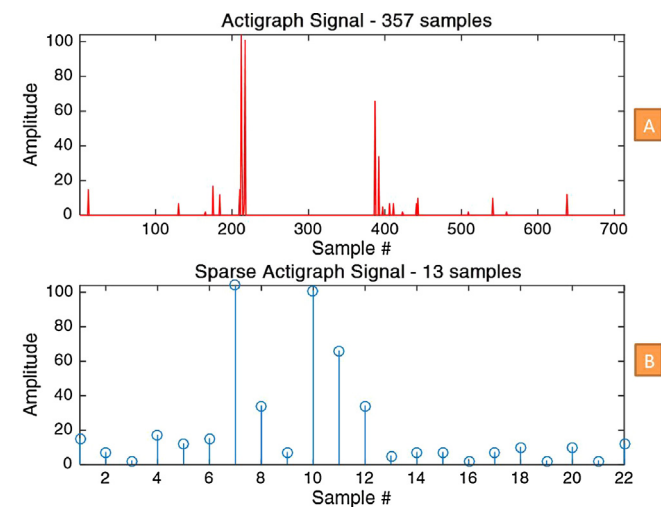


Fig. 17. (A) Actigraph signal, (B) sparse actigraph signal.

characteristic information about signal patterns, and could be later used for signal classification or signal reconstruction. Sparse techniques are also influencing hardware design including sensor development, by improving algorithm complexity, reducing memory and computational costs. A most apt example to illustrate this trend would be single pixel camera developed at Rice University [130] which imbibes compressed sensing algorithms for image capture and reconstruction without losing pixel-level information. Similar works are being done in audio signal processing in smart devices, security monitoring and the currently trending big data analytics. **One might speculate that sparse domain methods have evolved from time, frequency and joint time–frequency techniques. Although this is quite true, one must also keep in mind that sparse techniques could also be recombined with any other methodology for improved feature analysis of signals and would yield promising results (see Table 5 and Fig. 18).**

6. Significance of features for machine learning

Following our review on various feature extraction methods for biomedical signal analysis, we must also include the importance of implementing features as data representations in a machine learning paradigm. In concise words, the objective of a machine learning algorithm or tool is to categorize the behaviour of a signal emanat-

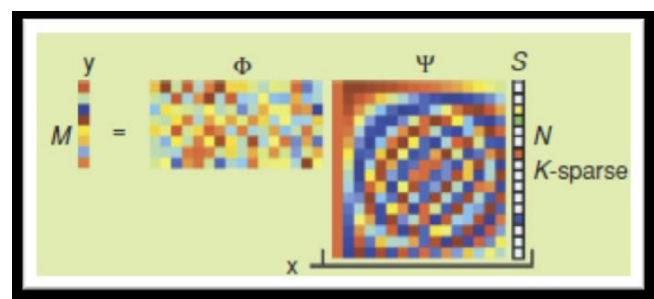


Fig. 18. Compressive sensing approach [121].

ing source by analysing the patterns generated in its data. The idea of pattern analysis is very much analogous to extracting features from a signal. Hence, it follows that in order to identify signal patterns correctly, it is imperative to select an appropriate machine learning algorithm, based on signal type and features extracted. This selection should also account for handling feature and signal robustness.

For example, assume that we have a pre-labelled database of about 100 ECG (electrocardiogram) signals which need to be segregated into normal and abnormal classes. Using a robust ECG suited algorithm, we extract about five features per signal, and generate testing and training feature sets for a machine learning tool. Now, we have a choice of applying either a support vector machine (SVM) or a simple Naive–Bayes algorithm to classify the 100 ECG signals. Ideally an SVM would be a better choice for this application, but practicality suggests otherwise. In this particular example, the computational cost and the overall cost of implementing a Naive–Bayes classifier is significantly lower than the SVM, along with the fact that for 5 feature classification, there would not be a significant difference in classification rates between the two. But say, if our dataset is now increased to about 10,000 ECG signals, each generating about say 20 features, then implementing a SVM would be more efficient, as it can handle higher dimensionality, scale and data size. Thus, in order to efficiently use features for signal classification, a machine learning algorithm must be selected based on the following criteria:

- Signal and feature dataset size.
- Type of biomedical signal to be analysed. For example, EEG (electroencephalogram) being more complex than ECG would need the aid of a complex machine learning tool such as SVM or deep learning algorithm, as opposed to using simple linear classifiers.

- Labelling of signal data.
- Implementation environment of the signal processing algorithm.

In cases where labelling of signals is tedious, machine learning techniques based on unsupervised learning could be implemented to help with annotation or labelling of the datasets. The main disconnect arises when features are extracted from short-duration segments of a biomedical signal. In most cases a clinical label is given to an entire signal, and generally there is lack of availability of segment-based or temporal region of interest (ROI) clinical labels. This opens up uncertainty in designing training stage of machine learning in which features extracted locally are given global labels, and in most cases there could be mis-matches. This is one area in which domain experts should be brought in enabling identifying ROIs from signals similar to what is done in medical image analysis. Generally, such a ROI identification from clinicians is not easy as unlike radiologists who help in identifying ROIs in images, domain experts are not well versed in interpreting the waveform aspect of biomedical signals. This is where domain experts can seek the assistance of an unsupervised machine learning approach such as clustering to assist in labelling signal segments, and this way proper labels are attributed to the features extracted from such segments. This is an open and exciting area of research that could provide true analysis of long-term biomedical signals for it to be a valuable attribute in the design of artificial intelligent systems for aiding clinical decision making.

The approach of machine learning-based labelling will become more important in the emerging context of wearable devices for biomedical signal data analysis. Deep learning is another emerging machine learning paradigm for handling vast collections of data [131,132]. The objective of a deep learning tool is to learn the representations of large datasets using multiple levels of non-linear, mathematical transformations, in a hierarchical fashion. The theory promotes the application of back propagation algorithms to define how a machine should adapt to represent data in each layer, using its representation from previous layer. Deep learning techniques have evolved from artificial neural network and convolutional network techniques, which have been extensively applied in speech recognition [133], computer vision [134,135] and biomedical signal and image processing areas [136,137], thus generating highly accurate results in pattern classification problems. [131,138,132].

Deep learning techniques are designed with the intention that minimal human engineering is required, and the machine itself generates multiple layers of non-linear transforms through learning previous representations. Therefore, each layer computes a non-linear transform of the input data, thus generating a set of values or coefficients which are parsed on to the next layer as new data for subsequent dimension reduction and transformation. The output of each layer could be considered as features directed towards the deep learning algorithm for pattern classification purposes [131,138]. As discussed in our review and illustrated in Fig. 19, the intention of signal-specific feature extraction tools is to extract characteristic information from signals, such that the signal behaviour is accurately represented. In case of deep learning, since each layer performs a blind non-linear transformation and dimension reduction, the features generated may not be sufficient or accurate enough to understand the signal or its source's behaviour, thereby rendering the deep learning procedure inefficient, especially in case of biomedical signal analysis, wherein understanding of the physiology of the disease and organs is vital towards decrypting the signal patterns. Although deep learning lacks the theoretical rigour in understanding its performances and outcomes, lots of research efforts are directed towards understanding the scientific basis for its capabilities towards feature representation.

7. Curse and blessing of dimensionality

While we debate and review which techniques could be useful in extracting robust features from multi-modal biomedical signals, we also need to consider the general aspects of curse and blessing of data/signal dimensionality. Choosing an algorithm which captures the best of these aspects should be our imperative goal in order to extract robust features.

In a feature extraction context, the curse of dimensionality means that our algorithm does not scale with the data, i.e. we might encounter Hughes' effect – infinite distribution from finite training data [139]. This could create a data compression problem, and may render the algorithm towards low level feature extraction. Extracting features is like finding a needle in the haystack. In order to do this, we might need to exploit the blessings of dimensionality by transforming our data/signal into higher dimensions wherein visualization, extracting hidden and temporal information becomes easier. Most biomedical signals are multi-variate and multi-modal in nature, and hence capturing multiple measurements at each observation, in higher dimensions is easier. Based on these concepts, the methods discussed in this review could also be regrouped into PC-based and Cloud-based categories. PC-based methods (from time and frequency domains) are usually static techniques, and are more suited to post-capture feature extraction; whereas the Cloud-based methods (from sparse and decomposition domains) have a real-time feature extraction capability by analysing the signal in burst mode. This being said each domain has given a tangible impact to the biomedical signal processing world:

- Time-domain methods have enabled the application of piecewise evaluation of signals coupled with high SNR for low level feature extraction.
- Frequency-domain gives us techniques which can preserve signal energy, retain phase components, understand periodic nature of signals, and are fairly simple to implement on hardware.
- Joint time-frequency plane can capture signal transients in 2D thus handling non-stationarity. TF methods also have inherent artefact removal, local and global feature extraction from signals.
- The currently trending signal decomposition methods generate maximally representable features with minimum redundancy of information, and can handle non-stationary and non-linear nature of signals.

8. Discussions, conclusions and future works

From this review we have observed that as we progress from time to signal decomposition domain, we see an evolutionary increase in signal data dimensions, methods which could exploit the orthogonality and correlation between signal values, possibilities of multi-domain feature combinations (local and global), and developments in multi-modal feature extraction tools with machine learning capabilities.

One must realize that real world biomedical signals are non-linear, non-stationary and could comprise of multi-modal components. Popular methods such as Autoregressive modeling, Cepstrum modeling, and Fourier and Wavelet analysis can handle non-stationarity through windowing approaches, but with certain limitations such as information loss, lower artifact filtering and low signal-to-noise and distortion-ratio (SNDR). We suggest that modern day feature extraction methods must be as intelligent and trainable as the pattern classifier itself. Ideally instead of windowing, it is recommended to handle real-time signals using a streaming or on-the-fly approach, i.e. extract features as the signal propagates through the source. This could be made possible by employing newer sparse and compressive sensing approaches

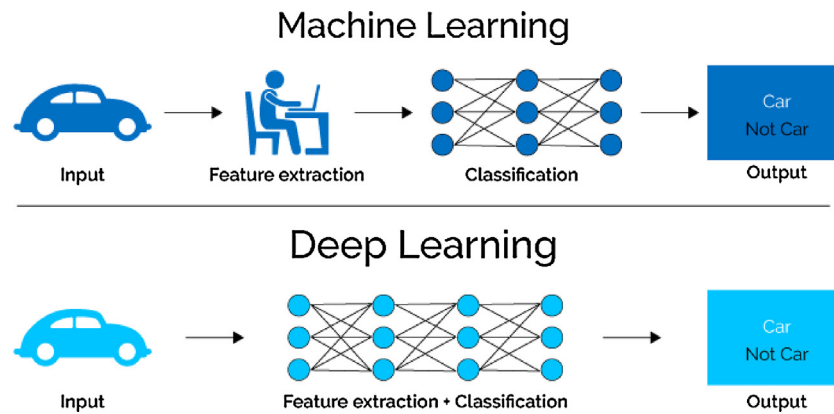


Fig. 19. Typical machine learning versus deep learning [132].

in combination with TF methods such as non-negative matrix factorization; or even a deep learning network constructed using a cascade of wavelet filters.

An intelligent feature extractor could possibly eliminate the need for a feature selection technique (such as mRMR – minimum redundancy maximum representation [140]), as this would happen inherently within the feature extractor. Additionally if the features are visually separable between signal classes, then employing a simple linear classifier could reduce system design constraints thus impacting the hardware design positively.

A practical feature extraction algorithm is one wherein we aim for robust feature generation coupled with low computation cost. In many applications these signal features could be coupled with clinical features and metadata for informed decision making in certain application domains. Our objective should be to create the highlighted block successfully – which can do data compression, trainable feature construction and selection. The output could be a feedback based, as it would improve the signal source. The reader must have noted in this review that we have emphasized a lot on the “robustness” of features. This is because, we want to ensure that however and whatever our signal source may be, the underlying feature extraction algorithm must be able to extract robust features every time. That is, the features extracted must be constrained/optimized to be robust and invariant to device or signal source changes.

Determination of the appropriate number of features that avoids over-fitting or under-fitting of the signal being analysed is an open ended problem. For example, even though in speech signal analysis it has been determined that 13 MFCC features are optimal to model a speech segment, in other areas this is still an open ended research. One way might be to pick the number of features as per the number of spectral peaks seen in that signal's spectrum. Although such an approach might work in certain contexts in which linearity and stationarity holds true, and in which case the spectral peaks correspond to the coefficients of an all-pole model (Autoregressive model); there are well established algorithms that can estimate an appropriate model order and estimate these model coefficients which in turn could be treated as signal features. There are a few non-linear modelling techniques available but are yet to be fully explored for appropriate feature extraction applications.

Through this study we have attempted to systematically review and determine the best combination of signal processing methods which could generate an intelligent feature extractor capable of: [i] robustness to artefacts, [ii] improving SNDR, [iii] handling non-linearity, [iv] handling non-stationarity, [v] assessing signal component variability, [vi] addressing higher dimensionality of feature space by compact or sparse feature generation, and most importantly [vii] generating a feature set which brings out maxi-

mum representation of the signal and helps pattern classification. From a hardware design and computational complexity perspective we should also include the following criterion: [i] built-in pre-processing and artefact removal, [ii] low power and memory consumption, [iii] real-time signal processing capability and [iv] inter-operability capabilities.

The authors would like to term that there needs to be a “science for feature extraction” in which determination of appropriate and characteristic features are explainable through information theory framework and other related ones. If such theories exist then feature extraction from biomedical signals will be a systematic area of study rather than adhoc and “blind” approaches adopted in many biomedical signal processing applications with empirical trial and error combinations. The area will further benefit if domain experts are involved as part of feature extraction design. This will reduce the gap that exists currently with intelligent systems in biomedical applications in which most feature extraction and machine learning implementations are seen as “black-boxes” by the domain expert or an end user. In conclusion, we would like to highlight that if we meet the said criteria, then the possibilities of developing a novel intelligent feature extractor are not far-fetched and could very well be a significant area of research and development in the biomedical signal processing world.

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References

- [1] Y. Athavale, Pattern classification of time-series signals using fisher kernels and support vector machines (Master's thesis), 2010, January, Available: <http://digital.library.ryerson.ca/islandora/object/RULA%3A1262>.
- [2] C. Guo, Z.-G. Hou, Z. Zeng (Eds.), *Advances in Neural Networks* ISSN 2013, Springer Berlin Heidelberg, 2013.
- [3] M. Kamel, A. Campilho (Eds.), *Image Analysis and Recognition*, Springer Berlin Heidelberg, 2009.
- [4] Y. Wu, in: *Advances in Computer, Communication, Control and Automation*, Springer Berlin Heidelberg, 2012.
- [5] D.-S. Huang, V. Bevilacqua, J.C. Figueroa, P. Premaratne (Eds.), *Intelligent Computing Theories*, Springer Berlin Heidelberg, 2013.
- [6] A.B. Geva, Feature extraction and state identification in biomedical signals using hierarchical fuzzy clustering, *Med. Biol. Eng. Comput.* 36 (5) (1998) 608–614.
- [7] S. Gibson, J.W. Judy, D. Markovic, Technology-aware algorithm design for neural spike detection, feature extraction, and dimensionality reduction, *IEEE Trans. Neural Syst. Rehabil. Eng.* 18 (5) (2010) 469–478.

- [8] C.J. James, C.W. Hesse, Independent component analysis for biomedical signals, *Physiol. Meas.* 26 (1) (2004) R15–R39.
- [9] D. Li, W. Pedrycz, N. Pizzi, Fuzzy wavelet packet based feature extraction method and its application to biomedical signal classification, *IEEE Trans. Biomed. Eng.* 52 (6) (2005) 1132–1139.
- [10] S. Preece, J. Goulermas, L. Kenney, D. Howard, A comparison of feature extraction methods for the classification of dynamic activities from accelerometer data, *IEEE Trans. Biomed. Eng.* 56 (3) (2009) 871–879.
- [11] Feature extraction in time and frequency domain: Kishore prahallad, <https://archive.org/details/FeatureExtractionInTimeAndFrequencyDomain> (accessed 14.09.16).
- [12] MITX, Time domain versus frequency domain analysis, <https://6002x.mitx.mit.edu/wiki/view/TimeDomainVersusFrequencyDomainAnalysis> (accessed 14.09.16).
- [13] P. Lutus, Signal processing workshop, <http://arachnoid.com/signal-processing/> (accessed 14.09.16).
- [14] D.H. Johnson, Statistical signal processing, <http://www.ece.rice.edu/dhj/courses/elec531/> (accessed 15.09.16).
- [15] N.R. Farnoud, M. Kolios, S. Krishnan, Ultrasound backscatter signal characterization and classification using autoregressive modeling and machine learning algorithms, *Engineering in Medicine and Biology Society, 2003. Proceedings of the 25th Annual International Conference of the IEEE*, vol. 3, September (2003) 2861–2864.
- [16] M. Shokrollahi, S. Krishnan, D. Jewell, B. Murray, *Autoregressive and Cepstral Analysis of Electromyogram in Rapid Movement Sleep*, Springer Berlin Heidelberg, 2010, pp. 1580–1583, http://dx.doi.org/10.1007/978-3-642-03882-2_419.
- [17] S. Krishnan, Adaptive signal processing techniques for analysis of knee joint vibroarthrographic signals (Ph.D. dissertation), 1999, June.
- [18] D. Hosseinzadeh, S. Krishnan, Gaussian mixture modeling of keystroke patterns for biometric applications, *IEEE Trans. Syst. Man Cybern. Part C (Appl. Rev.)* 38 (November (6)) (2008) 816–826.
- [19] H. Nallapareddy, S. Krishnan, M. Kolios, Parametric analysis of ultrasound backscatter signals for monitoring cancer cell structural changes during cancer treatment, *Can. Acoust.* 35 (2) (2007) 47–54, Available: <http://jcaa.caa-aca.ca/index.php/jcaa/article/view/1877>.
- [20] Y. Athavale, S. Krishnan, P. Hosseinzadeh, A. Guergachi, Identifying the potential for failure of businesses in the technology, pharmaceutical and banking sectors using kernel-based machine learning methods, *IEEE International Conference on Systems, Man and Cybernetics, 2009. SMC 2009*, October (2009) 1073–1077.
- [21] H. Asefi, B. Ghoraani, A. Ye, S. Krishnan, Audio scene analysis using parametric signal features, 2011 24th Canadian Conference on Electrical and Computer Engineering (CCECE), May (2011), pp. 000922–000925.
- [22] R.B. Randall, A history of cepstrum analysis and its application to mechanical problems, https://surveillance7.sciencesconf.org/conference/surveillance7/01_a_history_of_cepstrum_analysis_and_its_application_to_mechanical_problems.pdf (accessed 04.10.16).
- [23] P. Shokrollahi, S. Member, S. Krishnan, K. Umamathy, K. McConville, M.I. Boulos, D. Jewell, B.J. Murray, Computer-assisted method for quantifying sleep eye movements that reflects medication effects, 2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, September (2009) 1347–1350.
- [24] M. Shokrollahi, S. Krishnan, D. Kumar, S. Arjunan, Chin EMG analysis for REM sleep behavior disorders, 2012 ISSNIP Biosignals and Biorobotics Conference: Biosignals and Robotics for Better and Safer Living (BRC), January (2012) 1–4.
- [25] T.S. Tabatabaei, S. Krishnan, A. Guergachi, Emotion recognition using novel speech signal features, 2007 IEEE International Symposium on Circuits and Systems, May (2007) 345–348.
- [26] E. Shokrollahi, S. Krishnan, K. Nanthakumar, Transfer Function Estimation of the Right Ventricle of Canine Heart, Springer Berlin Heidelberg, Berlin, Heidelberg, 2010, pp. 1588–1591, http://dx.doi.org/10.1007/978-3-642-03882-2_421.
- [27] D. Hosseinzadeh, S. Krishnan, Combining vocal source and MFCC features for enhanced speaker recognition performance using GMMS, *IEEE 9th Workshop on Multimedia Signal Processing, MMSP 2007*, October (2007) 365–368.
- [28] K. Umamathy, B. Ghoraani, S. Krishnan, Audio signal processing using time-frequency approaches: coding, classification, fingerprinting, and watermarking, *EURASIP J. Adv. Signal Process* (2010, February), <http://dx.doi.org/10.1155/2010/451695>, 1:1–1:28.
- [29] K. Umamathy, S. Krishnan, A signal classification approach using time-width vs frequency band sub-energy distributions, in: in *Proceedings. (ICASSP '05). IEEE International Conference on Acoustics, Speech, and Signal Processing*, vol. 5, March 2005, pp. v/477–v/480 Vol. 5, 2005.
- [30] K.-P. Lin, W. Chang, QRS feature extraction using linear prediction, *IEEE Trans. Biomed. Eng.* 36 (10) (1989) 1050–1055.
- [31] J.P. Thiran, B. Macq, Morphological feature extraction for the classification of digital images of cancerous tissues, *IEEE Trans. Biomed. Eng.* 43 (October (10)) (1996) 1011–1020.
- [32] A. Aquino, M.E. Gegúndez-Arias, D. Marín, Detecting the optic disc boundary in digital fundus images using morphological, edge detection, and feature extraction techniques, *IEEE Trans. Med. Imaging* 29 (11) (2010) 1860–1869.
- [33] M.R.K. Mookiah, U.R. Acharya, R.J. Martis, C.K. Chua, C.M. Lim, E. Ng, A. Laude, Evolutionary algorithm based classifier parameter tuning for automatic diabetic retinopathy grading: a hybrid feature extraction approach, *Knowl. Based Syst.* 39 (2013) 9–22.
- [34] S. Zhou, J. Shi, J. Zhu, Y. Cai, R. Wang, Shearlet-based texture feature extraction for classification of breast tumor in ultrasound image, *Biomed. Signal Process. Control* 8 (6) (2013) 688–696.
- [35] Y. Athavale, S. Krishnan, D.D. Dopsa, A.G. Berneshawi, H. Nouraei, A. Raissi, B.J. Murray, M.I. Boulos, Advanced signal analysis for the detection of periodic limb movements from bilateral ankle actigraphy, *J. Sleep Res.* (2016, July).
- [36] Y. Athavale, S. Krishnan, A. Guergachi, Pattern classification of signals using fisher kernels, *Math. Probl. Eng.* 2012 (2012) 1–15.
- [37] T. Farooq, A. Guergachi, S. Krishnan, Chaotic time series prediction using knowledge based green's kernel and least-squares support vector machines, 2007 IEEE International Conference on Systems, Man and Cybernetics, October (2007) 373–378.
- [38] M. Sewell, The fisher kernel: a brief review, <http://www.cs.ucl.ac.uk/fileadmin/UCL-CS/research/Research.Notes/RN.11.06.pdf> (accessed 04.10.16).
- [39] Y. Tian, L. He, Z.Y. Li, W.I. Wu, W.Q. Zhang, J. Liu, Speaker verification using fisher kernels, 2014 9th International Symposium on Chinese Spoken Language Processing (ISCSLP), September (2014) 419–422.
- [40] V. Wan, S. Renals, Evaluation of kernel methods for speaker verification and identification, 2002 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), vol. 1, May (2002), pp. I-669–I-672.
- [41] V. Wan, S. Renals, Speaker verification using sequence discriminant support vector machines, *IEEE Trans. Speech Audio Process.* 13 (March (2)) (2005) 203–210.
- [42] The Fourier transform and its applications, <https://see.stanford.edu/Course/EE261> (accessed 15.09.16).
- [43] Frequency domain processing, <http://www.numerix-dsp.com/tutorials/DSP/FrequencyDomainProcessing.pdf> (accessed 15.09.16).
- [44] S.W. Smith, The discrete Fourier transform, <http://www.dspguide.com/ch8.htm> (accessed on 05.10.16).
- [45] L. Keselbrener, S. Akselrod, Selective discrete fourier transform algorithm for time-frequency analysis: method and application on simulated and cardiovascular signals, *IEEE Trans. Biomed. Eng.* 43 (August (8)) (1996) 789–802.
- [46] D. Hosseinzadeh, S. Krishnan, On the use of complementary spectral features for speaker recognition, *EURASIP J. Adv. Signal Process.* 2008 (January) (2008), <http://dx.doi.org/10.1155/2008/258184>, 46:1–46:10.
- [47] F. Jin, S. Krishnan, F. Sattar, Adventitious sounds identification and extraction using temporal spectral dominance-based features, *IEEE Trans. Biomed. Eng.* 58 (November (11)) (2011) 3078–3087.
- [48] A.P. Burgess, Towards a unified understanding of event-related changes in the EEG: the firefly model of synchronization through cross-frequency phase modulation, *PLoS ONE* 7 (9) (2012) e45630.
- [49] A. Ramalingam, S. Krishnan, Gaussian mixture modeling of short-time Fourier transform features for audio fingerprinting, *IEEE Trans. Inf. Forensics Secur.* 1 (December (4)) (2006) 457–463.
- [50] Lossy data compression: Jpeg, <http://cs.stanford.edu/people/eroberts/courses/soco/projects/data-compression/lossy/jpeg/dct.htm> (accessed 05.10.16).
- [51] S. Karpagachelvi, M. Arthanari, M. Sivakumar, ECG feature extraction techniques – a survey approach, *CoRR*, vol. abs/1005.0957, 2010 arXiv:1005.0957.
- [52] K. Umamathy, S. Mass, E. Sevaptisid, J. Asta, S. Krishnan, K. Nanthakumar, Spatiotemporal frequency analysis of ventricular fibrillation in explanted human hearts, *IEEE Trans. Biomed. Eng.* 56 (February (2)) (2009) 328–335.
- [53] S. Erkucuk, S. Krishnan, M. Zeytinoglu, A robust audio watermark representation based on linear chirps, *IEEE Trans. Multimedia* 8 (October (5)) (2006) 925–936.
- [54] L. Le, S. Krishnan, Time-frequency signal synthesis and its application in multimedia watermark detection, *EURASIP J. Adv. Signal Process.* 2006 (2006) 1–15.
- [55] L. Zhang, S. Krishnan, H. Ding, Modified spread spectrum audio watermarking algorithm, *Can. Acoust.* 32 (3) (2004) 142–143, Available: <http://jcaa.caa-aca.ca/index.php/jcaa/article/view/1668>.
- [56] Physionbank atm, <https://physionet.org/cgi-bin/atm/ATM> (accessed 05.10.16).
- [57] M. Klingspor, Hilbert transform: mathematical theory and applications to signal processing, <http://liu.diva-portal.org/smash/get/diva2:872439/FULLTEXT02.pdf> (accessed 05.10.16).
- [58] M. Unser, P.D. Tafti, *An Introduction to Sparse Stochastic Processes*, Cambridge University Press (Virtual Publishing), United Kingdom, 2014.
- [59] K. Umamathy, S. Krishnan, Time-frequency signal decompositions for audio and speech processing, *Can. Acoust.* 33 (3) (2005) 58–59, Available: <http://jcaa.caa-aca.ca/index.php/jcaa/article/view/1744>.
- [60] R.M. Rangayyan, S. Krishnan, Feature identification in the time-frequency plane by using the Hough-Radon transform, *Pattern Recognit.* 34 (6) (2001) 1147–1158, Available: <http://www.sciencedirect.com/science/article/pii/S003132030000073X>.
- [61] K. Umamathy, S. Krishnan, V. Parsa, D. Jamieson, Time-frequency modeling and classification of pathological voices, *Engineering in Medicine and Biology. 24th Annual Conference and the Annual Fall Meeting of the Biomedical Engineering Society EMBS/BMES Conference, 2002. Proceedings of the Second Joint vol. 1* (2002) 116–117.

- [62] K. Umapathy, S. Krishnan, V. Parsa, D.G. Jamieson, Discrimination of pathological voices using a time–frequency approach, *IEEE Trans. Biomed. Eng.* 52 (March (3)) (2005) 421–430.
- [63] S. Thayilchira, S. Krishnan, Detection of linear chirp and non-linear chirp interferences in a spread spectrum signal by using Hough–Radon transform, 2002 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), vol. 4, May (2002), p. IV-4181.
- [64] J. Yang, S. Krishnan, Wavelet packets-based speech enhancement for hearing aids application, *Can. Acoust.* 33 (3) (2005) 66–67, Available: <http://jcaa.caa-aca.ca/index.php/jcaa/article/view/1747>.
- [65] DSPRelated, The short-time Fourier transform | spectral audio signal processing, <https://www.dsprelated.com/freebooks/sasp/Short.Time.Fourier.Transform.html> (accessed 05.10.16).
- [66] Y. Shen, X. Li, N.-W. Ma, S. Krishnan, Parametric time–frequency analysis and its applications in music classification, *EURASIP J. Adv. Signal Process.* 2010 (2010) 1–9.
- [67] K. Umapathy, S. Krishnan, Perceptual coding of audio signals using adaptive time–frequency transform, *EURASIP J. Audio Speech Music Process.* 2007 (2007) 1–14.
- [68] J. Bonnel, A. Khademi, S. Krishnan, C. Ioana, Small bowel image classification using cross-co-occurrence matrices on wavelet domain, *Biomed. Signal Process. Control* 4 (1) (2009) 7–15 [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S1746809408000529>.
- [69] X. Li, S. Krishnan, N.W. Ma, A wavelet-PCA-based fingerprinting scheme for peer-to-peer video file sharing, *IEEE Trans. Inf. Forensics Secur.* 5 (September (3)) (2010) 365–373.
- [70] A. Khademi, S. Krishnan, A. Venetsanopoulos, Shift-invariant DWT for medical image classification, in: *Discrete Wavelet Transforms – Theory and Applications*, 2011.
- [71] Wavelet transform tutorial, <http://disp.ee.ntu.edu.tw/tutorial/WaveletTutorial.pdf> (accessed 15.09.16).
- [72] G. Chen, S. Krishnan, Small bowel image classification using dual tree complex wavelet-based cross co-occurrence features and canonical discriminant analysis, 2015 International Conference on Advances in Computing, Communications and Informatics (ICACCI), August (2015) 2174–2179.
- [73] K. Balasundaram, S. Masse, K. Nair, T. Farid, K. Nanthakumar, K. Umapathy, Wavelet-based features for characterizing ventricular arrhythmias in optimizing treatment options, 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2011) 969–972.
- [74] M. Shokrollahi, S. Krishnan, Non-negative matrix factorization and sparse representation for sleep signal classification, 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), July (2013) 4318–4321.
- [75] A. Khademi, S. Krishnan, Shift-invariant discrete wavelet transform analysis for retinal image classification, *Med. Biol. Eng. Comput.* 45 (12) (2007) 1211–1222, <http://dx.doi.org/10.1007/s11517-007-0273-z>.
- [76] S. Cai, S. Yang, F. Zheng, M. Lu, Y. Wu, S. Krishnan, Knee joint vibration signal analysis with matching pursuit decomposition and dynamic weighted classifier fusion, *Comput. Math. Methods Med.* 2013 (2013) 1–11.
- [77] A. Subasi, EEG signal classification using wavelet feature extraction and a mixture of expert model, *Expert Syst. Appl.* 32 (May (4)) (2007) 1084–1093, <http://dx.doi.org/10.1016/j.eswa.2006.02.005>.
- [78] D. Cvetkovic, E.D. beyli, I. Cosic, Wavelet transform feature extraction from human PPG, ECG, and EEG signal responses to ELF PEMF exposures: a pilot study, *Digit. Signal Process.* 18 (5) (2008) 861–874.
- [79] A. Khademi, S. Krishnan, Medical image texture analysis: a case study with small bowel, retinal and mammogram images, *CCECE 2008. Canadian Conference on Electrical and Computer Engineering*, May (2008), pp. 001 949–001 954.
- [80] K. Umapathy, S. Krishnan, S. Masse, X. Hu, P. Dorian, K. Nanthakumar, Optimizing cardiac resuscitation outcomes using wavelet analysis, 2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, September (2009) 6761–6764.
- [81] F.H. Foomany, K. Umapathy, L. Sugavaneswaran, S. Krishnan, S. Masse, T. Farid, K. Nair, P. Dorian, K. Nanthakumar, Wavelet-based markers of ventricular fibrillation in optimizing human cardiac resuscitation, 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2010) 2001–2004.
- [82] E. Afatmirni, K. Nanthakumar, S. Masse, K. Nair, T. Farid, S. Krishnan, P. Dorian, K. Umapathy, Predicting refrillation from pre-shock waveforms in optimizing cardiac resuscitation, 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2011) 251–254.
- [83] L. Sugavaneswaran, K. Umapathy, S. Krishnan, Exploiting the ambiguity domain for non-stationary biomedical signal classification, 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2010) 1934–1937.
- [84] R.E. Learned, A.S. Willsky, A wavelet packet approach to transient signal classification, *Appl. Comput. Harmon. Anal.* 2 (3) (1995) 265–278, Available: <http://www.sciencedirect.com/science/article/pii/S1063520385710196>.
- [85] S. Krishnan, R.M. Rangayyan, Automatic de-noising of knee-joint vibration signals using adaptive time–frequency representations, *Med. Biol. Eng. Comput.* 38 (1) (2000) 2–8.
- [86] B. Ghorani, S. Krishnan, Time–frequency matrix feature extraction and classification of environmental audio signals, *IEEE Trans. Audio Speech Lang. Process.* 19 (September (7)) (2011) 2197–2209.
- [87] B. Ghorani, S. Krishnan, R.J. Selvaraj, V.S. Chauhan, T wave alternans evaluation using adaptive time–frequency signal analysis and non-negative matrix factorization, *Med. Eng. Phys.* 33 (6) (2011) 700–711, Available: <http://www.sciencedirect.com/science/article/pii/S1350453311000117>.
- [88] S. Xie, S. Krishnan, Wavelet-based sparse functional linear model with applications to EEGs seizure detection and epilepsy diagnosis, *Med. Biol. Eng. Comput.* 51 (1–2) (2012) 49–60.
- [89] L. Sugavaneswaran, K. Umapathy, S. Krishnan, Ambiguity domain-based identification of altered gait pattern in ALS disorder, *J. Neural Eng.* 9 (4) (2012) 046004.
- [90] L. Sugavaneswaran, M. Balouchestani, K. Umapathy, S. Krishnan, Discriminative kernel learning in ambiguity domain, 2014 9th International Symposium on Communication Systems, Networks Digital Signal Processing (CSNDSP), July (2014) 261–265.
- [91] L. Sugavaneswaran, S. Xie, K. Umapathy, S. Krishnan, Time–frequency analysis via Ramanujan sums, *IEEE Signal Process. Lett.* 19 (June (6)) (2012) 352–355.
- [92] L. Sugavaneswaran, K. Umapathy, S. Krishnan, Discriminative time–frequency kernels for gait analysis for amyotrophic lateral sclerosis, 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2011) 2683–2686.
- [93] J.-J. Chen, R. Shiavi, Temporal feature extraction and clustering analysis of electromyographic linear envelopes in gait studies, *IEEE Trans. Biomed. Eng.* 37 (3) (1990) 295–302.
- [94] G. Bodenstern, H. Praetorius, Feature extraction from the electroencephalogram by adaptive segmentation, *Proc. IEEE* 65 (5) (1977) 642–652.
- [95] S. Xie, F. Jin, S. Krishnan, F. Sattar, Signal feature extraction by multi-scale PCA and its application to respiratory sound classification, *Med. Biol. Eng. Comput.* 50 (7) (2012) 759–768, <http://dx.doi.org/10.1007/s11517-012-0903-y>.
- [96] S. Xie, S. Krishnan, Dynamic principal component analysis with nonoverlapping moving window and its applications to epileptic eeg classification, *Sci. World J.* 2014 (2014) 1–10.
- [97] S. Xie, F. Jin, S. Krishnan, Sparse approximation of long-term biomedical signals for classification via dynamic PCA, 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2011) 7167–7170.
- [98] M. Shokrollahi, S. Krishnan, D. Jewell, B. Murray, Analysis of the electromyogram of rapid eye movement sleep using wavelet techniques, 2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, September (2009) 2659–2662.
- [99] A. Phinyomark, K. Limsakul, P. Phukpattaranont, A novel feature extraction for robust EMG pattern recognition, *CoRR*, vol. abs/0912.3973, 2009 arXiv:0912.3973.
- [100] H. Hassanpour, M. Mesbah, B. Boashash, Time–frequency feature extraction of newborn eeg seizure using SVD-based techniques, *EURASIP J. Adv. Signal Process.* 2004 (16) (2004) 898124, <http://dx.doi.org/10.1155/S1110865704406167>.
- [101] N. Shams, B. Ghorani, S. Krishnan, Audio feature clustering for hearing aid systems, *Science and Technology for Humanity (TIC-STH)*, 2009 IEEE Toronto International Conference, September (2009) 976–980.
- [102] S. Krishnan, R.M. Rangayyan, G.D. Bell, C.B. Frank, Adaptive time–frequency analysis of knee joint vibroarthrographic signals for noninvasive screening of articular cartilage pathology, *IEEE Trans. Biomed. Eng.* 47 (6) (2000) 773–783.
- [103] P.J. Loughlin, K.L. Davidson, Modified Cohen–Lee time–frequency distributions and instantaneous bandwidth of multicomponent signals, *IEEE Trans. Signal Process.* 49 (6) (2001) 1153–1165.
- [104] A. Ramalingam, S. Krishnan, Gaussian mixture modeling of short-time fourier transform features for audio fingerprinting, *IEEE Transactions on Information Forensics and Security* 1 (4) (2006) 457–463.
- [105] R.M. Rangayyan, S. Krishnan, Feature identification in the timefrequency plane by using the houghradon transform, *Pattern Recognit.* 34 (2001) 1147–1158.
- [106] M.G. Frei, I. Osorio, Intrinsic time-scale decomposition: time–frequency–energy analysis and real-time filtering of non-stationary signals, *Proc. R. Soc. Lond. A: Math. Phys. Eng. Sci.* 463 (2078) (2007) 321–342, Available: <http://rspa.royalsocietypublishing.org/content/463/2078/321>.
- [107] M. Kaleem, B. Ghorani, A. Guergachi, S. Krishnan, Pathological speech signal analysis and classification using empirical mode decomposition, *Med. Biol. Eng. Comput.* 51 (7) (2013) 811–821.
- [108] J. Mairal, F. Bach, J. Ponce, Task-driven dictionary learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 34 (April (4)) (2012) 791–804.
- [109] M.F. Kaleem, A. Guergachi, S. Krishnan, A variation of empirical mode decomposition with intelligent peak selection in short time windows, 2013 IEEE International Conference on Acoustics, Speech and Signal Processing, May (2013) 5627–5631.
- [110] M. Kaleem, A. Guergachi, S. Krishnan, Eeg seizure detection and epilepsy diagnosis using a novel variation of empirical mode decomposition, 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), July (2013) 4314–4317.

- [111] M. Kaleem, A. Guergachi, S. Krishnan, Application of a variation of empirical mode decomposition and teager energy operator to EEG signals for mental task classification, 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), July (2013) 965–968.
- [112] M.F. Kaleem, B. Ghorani, A. Guergachi, S. Krishnan, Telephone-quality pathological speech classification using empirical mode decomposition, 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2011) 7095–7098.
- [113] M.F. Kaleem, A. Guergachi, S. Krishnan, A.E. Cetin, Using a variation of empirical mode decomposition to remove noise from signals, 2011 21st International Conference on Noise and Fluctuations (ICNF), June (2011) 123–126.
- [114] M. Kaleem, A. Guergachi, S. Krishnan, Empirical mode decomposition based sparse dictionary learning with application to signal classification, Digital Signal Processing and Signal Processing Education Meeting (DSP/SPE), 2013 IEEE, August (2013) 18–23.
- [115] M. Shokrollahi, S. Krishnan, Sleep emg analysis using sparse signal representation and classification, 2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, August (2012) 3480–3483.
- [116] M. Balouchestani, K. Sugavaneswaran, S. Krishnan, Advanced k -means clustering algorithm for large ecg data sets based on k -SVD approach, 2014 9th International Symposium on Communication Systems, Networks Digital Signal Processing (CSNDSP), July (2014) 177–182.
- [117] M. Balouchestani, K. Raahemifar, S. Krishnan, Low sampling rate algorithm for wireless ECG systems based on compressed sensing theory, Signal Image Video Process. 9 (3) (2013) 527–533.
- [118] A. Cichocki, D. Mandic, L.D. Lathauwer, G. Zhou, Q. Zhao, C. Caiafa, H.A. Phan, Tensor decompositions for signal processing applications: from two-way to multiway component analysis, IEEE Signal Process. Mag. 32 (March (2)) (2015) 145–163.
- [119] I. Tosic, P. Frossard, Dictionary learning for stereo image representation, IEEE Trans. Image Process. 20 (April (4)) (2011) 921–934.
- [120] I. Tosic, P. Frossard, Dictionary learning, IEEE Signal Process. Mag. 28 (March (2)) (2011) 27–38.
- [121] E.J. Candes, M.B. Wakin, An introduction to compressive sampling, IEEE Signal Process. Mag. 25 (March (2)) (2008) 21–30.
- [122] J.K. Pant, S. Krishnan, Compressive sensing of electrocardiogram signals by promoting sparsity on the second-order difference and by using dictionary learning, IEEE Trans. Biomed. Circuits Syst. 8 (April (2)) (2014) 293–302.
- [123] J. Andn, S. Mallat, Multiscale scattering for audio classification.
- [124] J.K. Pant, S. Krishnan, Foot gait time series estimation based on support vector machine, in: 2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, Institute of Electrical & Electronics Engineers (IEEE), 2014.
- [125] J.K. Pant, S. Krishnan, Reconstruction of ecg signals for compressive sensing by promoting sparsity on the gradient, 2013 IEEE International Conference on Acoustics, Speech and Signal Processing (2013).
- [126] J.K. Pant, S. Krishnan, Compressive sensing of ECG signals based on mixed pseudonorm of the first- and second-order differences, in: 2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Institute of Electrical & Electronics Engineers (IEEE), 2014.
- [127] J.K. Pant, S. Krishnan, Compressive sensing of foot gait signals and its application for the estimation of clinically relevant time series, IEEE Trans. Biomed. Eng. 63 (7) (2016) 1401–1415.
- [128] K. Mitra, A. Veeraraghavan, A.C. Sankaranarayanan, R.G. Baraniuk, Toward compressive camera networks, Computer 47 (May (5)) (2014) 52–59.
- [129] C. Li, T. Sun, K.F. Kelly, Y. Zhang, A compressive sensing and unmixing scheme for hyperspectral data processing, IEEE Trans. Image Process. 21 (March (3)) (2012) 1200–1210.
- [130] M.F. Duarte, M.A. Davenport, D. Takhar, J.N. Laska, T. Sun, K.F. Kelly, R.G. Baraniuk, Single-pixel imaging via compressive sampling, IEEE Signal Process. Mag. 25 (March (2)) (2008) 83–91.
- [131] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, Nature 521 (7553) (2015) 436–444.
- [132] Log analytics with deep learning and machine learning, <https://www.upwork.com/hiring/for-clients/log-analytics-deep-learning-machine-learning/> (accessed 20.09.17).
- [133] G. Hinton, L. Deng, D. Yu, G.E. Dahl, A.-r. Mohamed, N. Jaitly, A. Senior, V. Vanhoucke, P. Nguyen, T.N. Sainath, et al., Deep neural networks for acoustic modeling in speech recognition: the shared views of four research groups, IEEE Signal Process. Mag. 29 (6) (2012) 82–97.
- [134] A. Krizhevsky, I. Sutskever, G.E. Hinton, Imagenet classification with deep convolutional neural networks, Advances in neural information processing systems (2012) 1097–1105.
- [135] C. Farabet, C. Couprie, L. Najman, Y. LeCun, Learning hierarchical features for scene labeling, IEEE Trans. Pattern Anal. Mach. Intell. 35 (8) (2013) 1915–1929.
- [136] J.J. Tompson, A. Jain, Y. LeCun, C. Bregler, Joint training of a convolutional network and a graphical model for human pose estimation, Advances in Neural Information Processing Systems (2014) 1799–1807.
- [137] M.K. Leung, H.Y. Xiong, L.J. Lee, B.J. Frey, Deep learning of the tissue-regulated splicing code, Bioinformatics 30 (12) (2014) i121–i129.
- [138] D. Erhan, Y. Bengio, A. Courville, P.-A. Manzagol, P. Vincent, S. Bengio, Why does unsupervised pre-training help deep learning? J. Mach. Learn. Res. 11 (February) (2010) 625–660.
- [139] G. Hughes, On the mean accuracy of statistical pattern recognizers, IEEE Trans. Inf. Theory 14 (January (1)) (1968) 55–63.
- [140] Peng, MRMR feature selection site, <http://penglab.janelia.org/proj/mRMR/> (accessed 11.10.16).



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